Roscoe V2 - Change plan

This is the working list of every change we decide to make: physical rework on the current boards, CPLD source changes, expansion-bus definitions, and items for the next PCB. Changes are grouped into change sets (CS). Each set says what to wire, what to edit in which CPLD file, how to program and test it, and how to roll it back. Background and analysis are in TheoryOfOperations.md.

Nothing in this file has been programmed yet. CS1's CUPL was written against BUSCTRL rev 29 and DEVMAP rev 20/21 and now compiles (see the change log). CS2's is written against CS1's revisions plus INTCTRL rev 16 and still needs a WinCUPL compile, with the fit report checked for logic levels as noted in each step.

Before opening an edited .PLD in WinCUPL, run `powershell -ExecutionPolicy Bypass -File plds\check-wincupl-open.ps1`. WinCUPL's editor cannot open some files that compile cleanly and reports them as "doesn't exist. New File assumed". The script runs the editor's parser and names the line it would stop on.

Status values: Proposed -> Agreed -> Implemented (source edited) -> Programmed -> Verified.

Set	Title	Status
CS0	Pull-up rework already on the boards	Done on current boards; fix on next PCB
CS1	Device-cycle acknowledge (replaces `nDEV_WAIT`), fast VMEM window with card hold-off, bus timeout, refresh during long cycles	**Implemented** (sources edited 2026-10-18: `BUSCTRL2.PLD` rev 30, no-card build; `DEVMAP2.PLD` rev 22; INTCTRL unchanged). Both compile and fit in WinCUPL with every pin kept (DEVMAP 36/128 cells, BUSCTRL 79/128); not yet programmed. Backups: `BUSCTRL2_REV29.PLD`, `DEVMAP2_REV20.PLD`.
CS2	IDE timing on both ports, /IOR window, interrupts, pull-downs, reset length	**Agreed (2.11)**; implement after CS1 is verified. P7 (buffer swap) proposed.
CS3	IDE IORDY flow control (base window)	**Optional**: only if cards still fail after CS2 and the 3.4 check shows IORDY activity

CS0 - Rework already on the boards

#	Change	Current boards	Next PCB
0.1	Pull-ups on 68030 `/CDIS` (H12) and `/MMUDIS` (F13)	One wire joins PGA pins H12 and F13 on the underside and runs to J23 ("5VPU"),	10K pull-ups on the CPU sheet (H11 and F11 next to them are VCC)

#	Change	Current boards	Next PCB
		the header for the two spare 10K pull-ups R44 (J23 pin 1, EXTRAPU1) and R45 (pin 2, EXTRAPU2).	
0.1a	/CBACK (J1)	**Not wired**: still floating. Harmless today, because the 68030 ignores CBACK in every cycle terminated by DSACKx or BERR (UM 7.2.8), and Roscoe never asserts STERM.	route it to a CPLD, or to GND through a jumper (J3 next to it is GND), before any STERM or burst cycles (SRAMBurstProposal.md 4.1)
0.2	Pull-up on RTC /IRQ (U33 pin 19, vINT1A)	A wire from U33 pin 19 to the other J23 pin, so /IRQ uses the other spare pull-up. Confirmed on the photos.	10K pull-up
0.3	Filter capacitor on the reset button: from the reset-button net (PWR_RSTBUTTON: switch U66, J17 pin 1, ATtiny U65 pin 12 RESET_BUTTON) to VCC, using J301 pin 2 as the VCC point	added because the button input was noisy; value not recorded	Put the capacitor on the board next to U65 (button net to VCC), plus a 10K pull-up. The net has no resistor today, so it relies on the ATtiny's weak internal pull-up. Going to VCC rather than GND is right for an active-low button: at power-up the cap pulls the line high, so it never looks like a press.

The pin numbers come from the underside photos (IMG_0019, IMG_0022, IMG_0023), matched against the netlist and the PCB layout (V2PCBTop.pdf, V2PCBBottom.pdf). The photos show the board from below, which mirrors the layout. On the CPU, H12 and F13 are the only two pins that fit: both are unconnected in the netlist. Details: TheoryOfOperations.md, Section 15.

CS1 - Device-cycle acknowledge

1.1 Problem

Today BUSCTRL acknowledges the 8-, 16- and 32-bit device regions combinationally, about two CPLD levels after AS. DEVMAP's `nDEV_WAIT` can only arrive later: it waits for BUSCTRL's AS-qualified region select, does its own decode, clocks a flop, and drives an output. By then the CPU has already sampled DSACK. Pulling DSACK back up after that cannot add a wait state, however fast the pull-up is (the boards have 1K on R51/R52). The result is that no device can ever get a wait state, IDE included, and expansion cards would have no way to stretch a cycle.

1.2 Design

Turn "wait" into **acknowledge**: a device cycle ends only when the device side says so.

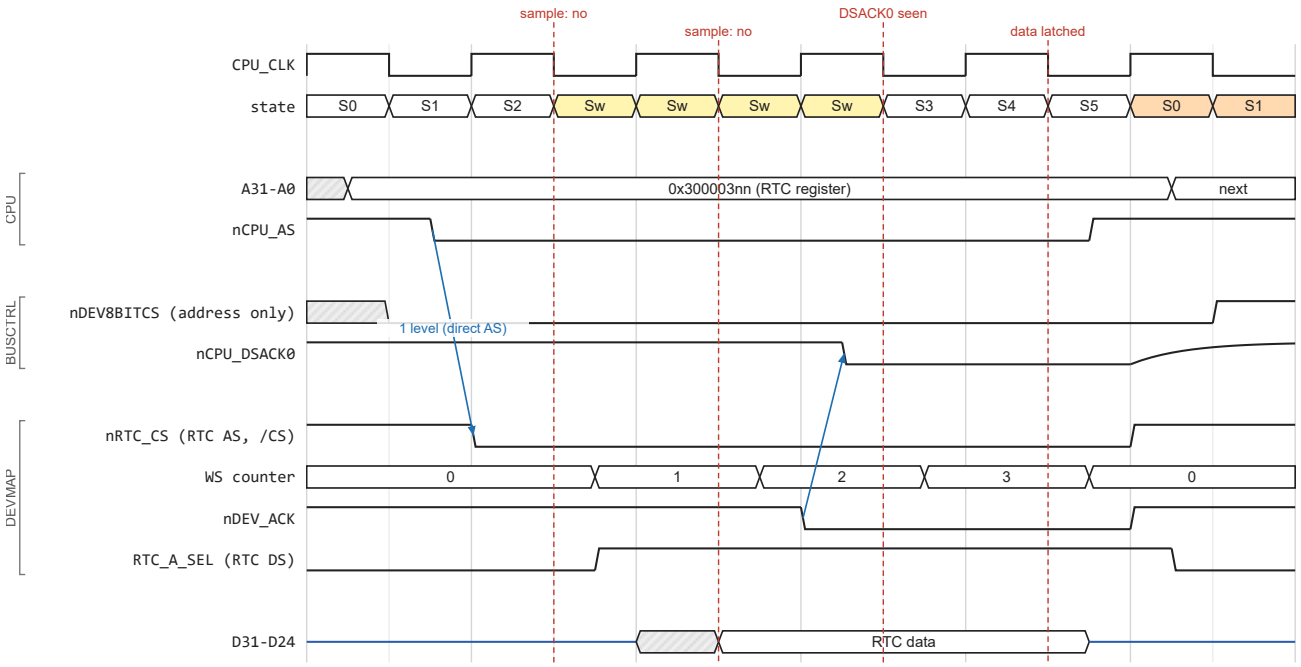
1. **BUSCTRL** asserts DSACK for a device region only while DEVMAP drives `nDEV_ACK` low. It uses the existing `nDEV_WAIT` trace (DEVMAP pin 6 to BUSCTRL dedicated input 90) with the meaning inverted. Once acknowledged, the acknowledge is held until AS negates, so DSACK is never retracted.
2. **DEVMAP** decides when each device is ready:
    - fast devices (UARTs, INTCTRL, displays, LEDs, PTC, empty on-board windows) acknowledge immediately: 0 wait states;
    - the RTC and IDE acknowledge after a fixed number of clocks, from a falling-edge counter that starts with AS;
    - expansion cards acknowledge themselves through the connector line `nEXP_ACK` (today's `nEXP_DEV_WAIT`).
3. **DEVMAP gets AS directly** (one rework wire), and BUSCTRL sends it the region decodes **without** AS. Those are valid well before AS, so DEVMAP's selects and acknowledges follow AS by a single CPLD level instead of two. This is what makes 0 wait states reachable at 16 MHz.
4. **The VMEM window is acknowledged by BUSCTRL itself**, at 0 wait states, like the SRAM. A VGA card can hold the CPU off while its display fetch is using the memory, through a busy/grant pair of connector lines. This is the 0-wait-state path for a VGA card (1.4).
5. **Bus timeout:** with "wait" now the default, a cycle that nobody acknowledges must not hang the machine. BUSCTRL asserts BERR after about 15 periods of CPU_CLK/16. This also fixes today's hangs: the 0x6/0x7 region, `BKPT`, and FPU instructions with no FPU fitted. VMEM cycles are excluded, because a hold-off may legitimately run longer.
6. **Refresh during long cycles:** today a DRAM refresh can only start between bus cycles, so a long hold-off or a slow card cycle would block it. BUSCTRL now also starts a refresh during any cycle that cannot be a DRAM cycle (1.4, B8).

Who ends each cycle after CS1:

Address space	Ended by	Wait states
SRAM, boot flash, big-flash region, interrupt acknowledge	BUSCTRL decode	0 (unchanged)
DRAM	BUSCTRL wait-state flop	1 (unchanged)
VMEM fast window 0x5000_0000-0x501F_FFFF	BUSCTRL grant; the card may hold it off	0, plus 1 per clock of hold-off
UART A/B, INTCTRL, 7-segment, LEDs, PTC, empty on-board windows	DEVMAP, immediately	0 (worst case 1 at 16 MHz, see 1.3)
RTC 0x3000_03xx	DEVMAP, after count 2	2
IDE1/IDE2 0x4000_00xx-0x4000_03xx	DEVMAP, after count 3	3 (interim, CS2 tunes it)
Card space (1.6) and the VIO window 0x3000_05xx	card via nEXP_ACK	card-defined, at least 1
Anything nobody acknowledges (0x6/0x7, empty card space, BKPT, missing FPU), except VMEM	BUSCTRL timeout	BERR after 224-240 clocks
0x2xxx_xxxx	BUSCTRL	BERR at once (unchanged)

The "on-board area" of the 8- and 16-bit regions is A27-A24 = 0 (0x3000_0000-0x30FF_FFFF and 0x4000_0000-0x40FF_FFFF). This matches the windows DEVMAP already decodes, and the DEVMAP comments' plan to put cards at 0x31.., 0x41.. and 0x51...

PROPOSED (changes.md CS1): RTC read with the device acknowledge - 2 wait states



Proposed. DEVMAP gets AS directly (rework wire) and BUSCTRL's region select early (address only), so device selects follow AS by one CPLD level. A falling-edge counter starts with AS. nDEV_ACK stays high (not ready) until the selected device's count is reached; for the RTC that is count 2, giving 2 wait states. UART, INTCTRL, LEDs and PTC acknowledge at count 0 (no wait states). Cards acknowledge through nEXP_ACK. BUSCTRL turns nDEV_ACK into DSACK and never retracts it.

1.3 Timing: what can run at 0 wait states

For 0 wait states, DSACK must be asserted by the falling edge that ends S2, less the 4 ns setup (#47A). AS can arrive as late as 20 ns into S1 (#9). The time available from AS to DSACK is therefore **T - 24 ns**: 59 ns at 12 MHz, 38.5 ns at 16 MHz, 26 ns at 20 MHz. Each missed sample costs exactly one wait state; nothing breaks.

Worst-case paths, with a -10 CPLD with slow-slew outputs (14 ns per level to a pin, 23 ns for a 2-level DSACK):

Path from AS to DSACK	Delay	12 MHz	16 MHz	20 MHz
BUSCTRL only: SRAM, flash	14-23 ns	0 WS	0 WS	0 WS
VMEM grant: AS must meet the 7 ns setup at the S2 rising edge, then about 20 ns from that edge to DSACK	(registered)	0 WS	0 WS (4 ns margin on AS)	0-1 WS
On-board device, today's wiring (AS-qualified region, then DEVMAP, then BUSCTRL)	42-51 ns	0 WS	1 WS	1 WS
On-board device with the AS wire (CS1)	28-37 ns	0 WS	0 WS (1.5-10 ns margin)	1 WS
Card handshake (U53, card logic, DEVMAP, BUSCTRL)	about 57-62 ns	0-1 WS	1 WS	1 WS

The AS wire is what buys 0-wait-state devices at 16 MHz. Without it CS1 still works, with every device cycle 1 wait state longer at 16 MHz.

Keep DSACK single-level. After compiling, check the fit report: `nCPU_DSACK0/1` should not depend on `XXL_` nodes. Keep the two DSACKs' 32-bit terms at the same depth, or 32-bit cycles can break the 68030's DSACK skew limit (#31A). B7 records how rev 30 achieves this.

1.4 VGA memory: 0 wait states, with a hold-off for the card

A per-cycle acknowledge from the card **cannot** give 0 wait states at 16 MHz. The round trip (buffered AS to the card, card decode, `nEXP_ACK` back through DEVMAP and BUSCTRL) is about 57-62 ns, and the window is 38.5 ns. It would only just make it at 12 MHz.

So for the video-memory window the motherboard ends the cycle itself, and the card only says **in advance** whether it can take a CPU access. The card's memory is dual-use: one SRAM array shared by the CPU and the display fetch. While the fetch has it, CPU accesses must wait, sometimes for many clocks. Two direct BUSCTRL pins that already reach the connector, both unused today, carry the handshake:

Pin	Signal	Direction	Meaning
B26	`nEXP_VMEM_BUSY`	card to BUSCTRL pin 28	Low: the card is using the memory, or is about to. CPU accesses wait. 4.7K pull-up on the board (P3), so with no card it is never busy.
B30	`nEXP_VMEM_GNT`	BUSCTRL pin 29 to card	Low: the CPU's current VMEM cycle has been granted. The card must complete it, and must not start a video access until GNT goes high again.

BUSCTRL. All decisions are taken on rising CPU_CLK edges. The first is the edge that starts S2 of a VMEM cycle. If the cycle is not granted there, BUSCTRL tries again on every later rising edge. It grants when `nEXP_VMEM_BUSY` is high at that edge. A grant drives `nEXP_VMEM_GNT` low and asserts DSACK0 and DSACK1 (a 32-bit port), and both stay asserted until AS negates, so an acknowledge is never withdrawn. Every decision edge at which BUSY is low adds one wait state.

The card. Its arbiter runs on the rising edges of `EBUS_CPU_CLK` (B27). That is a separate U1 section driven by the same oscillator as CPU_CLK, so the card and BUSCTRL decide on the same edges:

1. Between video accesses, the memory sits on the CPU path (CPU address, select and data buffers). A granted CPU cycle simply finds it there.
2. To start a video access, pull BUSY low on a rising edge. Sample GNT on each following rising edge, and start the access on the first one at which GNT is high. If GNT is low, the CPU was granted on the very edge you asked, or earlier: leave the memory on the CPU path, let that cycle finish, and wait for GNT to rise. That costs the video side at most 3 clocks.
3. While BUSY is low BUSCTRL grants nothing, so the memory is yours for as long as you keep BUSY low.

4. To give the memory back, drive BUSY high on a rising edge. You may keep using the memory until the next rising edge, the earliest one on which BUSCTRL can grant. From that edge on, the memory must be on the CPU path.
5. Reads: whenever the CPU is reading the window (`nEXP_VMEM_CS` low, `EXP_RW` high), drive `EXP_D` from the memory and hold `nEXP_DATA_EN` low until the select goes away. Before the grant the CPU ignores the data, so it does not matter that it may be stale.
6. Writes: change the memory only while GNT is low (timing below).
7. Keep each hold-off bounded: BUSY low for no more than about one scan line (32 us) at a time. VMEM cycles are excluded from the bus timeout, so a card that never releases BUSY hangs the machine.

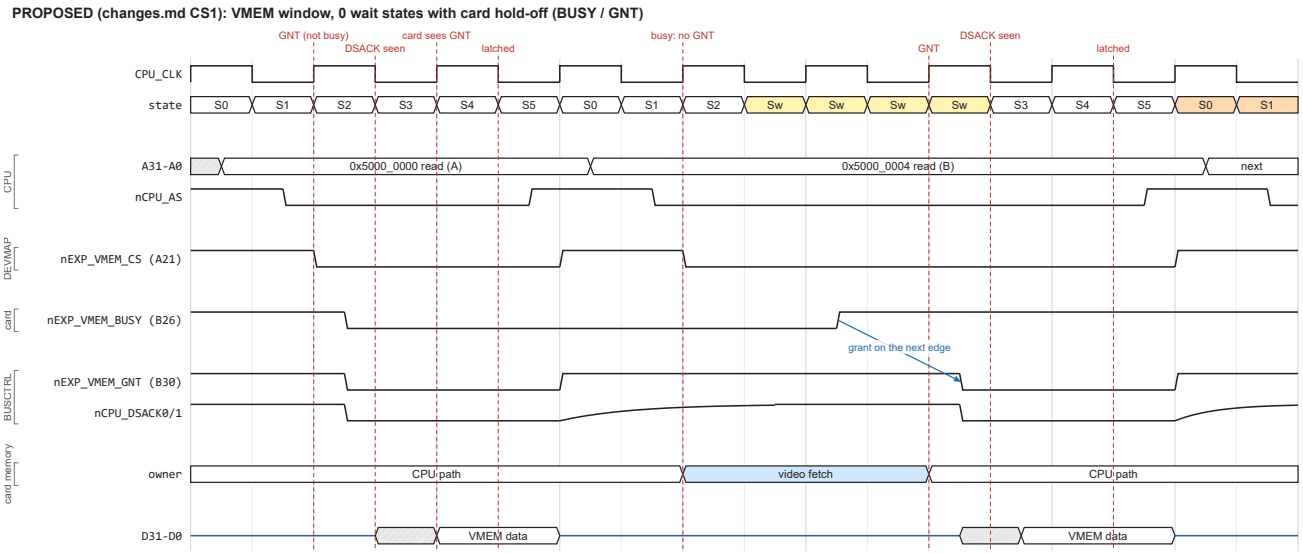
Timing. T is the CPU clock period, and edge k is the rising edge on which BUSCTRL grants.

Requirement	Formula	12 MHz	16 MHz	20 MHz
`nEXP_VMEM_BUSY` set-up before a rising edge, at BUSCTRL pin 28	7 ns	7 ns	7 ns	7 ns
`nEXP_VMEM_GNT` valid after edge k (registered pin, slow slew)	about 10 ns	10 ns	10 ns	10 ns
Read data valid at the connector, after edge k	1.5T - 12 ns	113 ns	82 ns	63 ns
The same for a 0-wait-state cycle, measured from `nEXP_VMEM_CS` falling	2T - 50 ns	117 ns	75 ns	50 ns
Earliest AS negation, which ends a write, after edge k	1.5T	125 ns	94 ns	75 ns

Write data is valid at the connector from about 35 ns into S2 (#23 plus the 245) until AS negates, with only 10 ns of hold after that (#25). A synchronous card can capture address, data and byte enables on the first rising edge at which it sees GNT low (edge k+1) and write the memory from that copy. Or it can strobe the memory directly between edge k+1 and the next falling edge. Either is easy for 10-15 ns SRAM.

Effects on the rest of the board.

- Bus timeout: VMEM cycles are excluded (B5), because a hold-off can legitimately last longer than the 14-20 us timeout.
- DRAM refresh: today a refresh can only start between bus cycles. A CPU held off for most of every scan line would then get about one refresh per 32 us line, and the DRAM needs one every 15.6 us on average. B8 lets a refresh start during any cycle that cannot be a DRAM cycle, so a stalled VMEM or card cycle no longer delays it.
- Interrupts wait until the stalled cycle ends, so interrupt latency grows by the length of the hold-off.



Proposed. Drawn at 16 MHz (1 slot = 15.6 ns). BUSCTRL decides the VMEM grant on rising CPU_CLK edges, starting with the edge that begins S2: it grants if the card's nEXP_VMEM_BUSY is high, then holds GNT and DSACK until AS negates, so an acknowledge is never withdrawn. Cycle A: the card asks for the memory (BUSY low) on the same edge that grants the CPU. It sees GNT on the next edge, so it leaves the memory on the CPU path and waits for GNT to clear; here it starts its fetch on the S2 edge of cycle B (worst case the video side waits 3 clocks). Cycle B: BUSY is low at every decision edge, so the CPU collects wait states while the card fetches. The card releases BUSY one clock before its fetch ends; BUSCTRL grants on the next edge and the read completes with 2 wait states. With no card fitted the pull-up keeps BUSY high and every VMEM cycle has 0 wait states.

Why decide on clock edges. If BUSCTRL simply gated a combinational acknowledge with BUSY, a card raising BUSY could collide with a cycle acknowledged a few nanoseconds earlier, and the CPU would read while the display fetch owned the memory. Deciding on clock edges, and handing the result back on GNT, gives both sides the same answer for every cycle. An earlier draft let the card raise BUSY only while AS was high. That avoids the collision, but the card would have to see AS high on the single rising edge between back-to-back CPU cycles: only a few nanoseconds of margin once the U53 buffer is counted. A CPU streaming to VMEM could also starve the display.

The window is 2 MB, 0x5000_0000-0x501F_FFFF (agreed). That covers the display and font areas of the current memory map and keeps the NIC at 0x5020_0000 in handshake space. Today's `nEXP_VMEM_CS` covers 1 MB; the size is one line in each CPLD.

1.5 Physical changes

#	Change	When	Notes
P1	**Wire `nCPU_AS` to DEVMAP pin 35**: logic header U70 pin 4 (`nCPU_AS`) to J28 pin 3 (DEVMAP pin 35, today `TESTP0`)	required for CS1	Short wire, ideally twisted with a ground. Any `nCPU_AS` point closer to U59 also works. Pin 35 is chosen because J28 brings it to a header, so no fine-pitch soldering. **Only fit it after DEVMAP rev 22 is programmed**: the old DEVMAP drives pin 35 as an output.
P2	R10 (`nEXP_DEV_WAIT`, becoming `nEXP_ACK`) 10K -> 1K	before the first card	Lets an open-drain card acknowledge release fast enough for the next cycle

#	Change	When	Notes
P3	**4.7K pull-up to +5V on `nEXP_VMEM_BUSY`** (connector B26). The easiest point is logic header U69 pin 3 (CLK1, `EXP_BC_CPLD1`), to any +5V point; U69 pin 1 is not connected.	**Deferred (decision 9):** fit it with the VGA card, when BUSCTRL switches to the hold-off build (B4a). The no-card build (B4b) used for CS1 ignores B26.	A BUSCTRL that uses the hold-off reads B26. With no card the pull-up keeps it high, so VMEM cycles run at 0 wait states. Without it the pin keeper holds whatever level the pin last had, and a low level would hang every VMEM access. Harmless on the old BUSCTRL, which drives B26 low: 1 mA.

The DSACK pull-ups need no change. R51/R52 are 1K on the boards (the schematic says 10K), which returns DSACK above 2.0 V in roughly 15-25 ns. The schematic gets corrected on the next PCB (1.13).

All three rework points were checked against the netlist (2026-10-18):

- `nCPU_AS` is on U70 pin 4.
- J28 pin 3 is DEVMAP pin 35.
- `EXP_BC_CPLD1` is BUSCTRL pin 28, connector B26 on all three slots, and U69 pin 3.

1.6 Expansion bus after CS1

The card acknowledge, `nEXP_ACK`, is connector pin **A26**. It is today's `nEXP_DEV_WAIT` net (sheet 14): pulled up by R10 and wired straight to DEVMAP pin 32. CS1 changes only its meaning (low now means "done" rather than "wait"); there is no new wiring.

Pin	Today	After CS1
A26	`nEXP_DEV_WAIT`: card pulls low to add wait states (has no effect today)	**`nEXP_ACK`**: card pulls low when its cycle may end. Required in card space.
B19, A20, B20	`nEXP_8BDEV`, `nEXP_16BDEV`, `nEXP_32BDEV`, AS-qualified region selects	unchanged meaning (now one CPLD level after AS)
A21	`nEXP_VMEM_CS`, 1 MB	2 MB fast window. The motherboard ends the cycle at 0 wait states unless the card holds it off (1.4).
A30	`nEXP_VIO_CS` (0x3000_05xx)	unchanged select; the window is now card space, so the card must acknowledge
B26	`EXP_BC_CPLD1`, driven 0	**`nEXP_VMEM_BUSY`**, card to board, 4.7K pull-up (P3)
B30	`EXP_BC_CPLD2`, driven 0	**`nEXP_VMEM_GNT`**, board to card

Address space for cards:

Space	Range	Ended by
8-bit card space	0x3100_0000-0x3FFF_FFFF, plus the VIO window 0x3000_0500-0x3000_05FF	nEXP_ACK
16-bit card space	0x4100_0000-0x4FFF_FFFF	nEXP_ACK
32-bit fast window	0x5000_0000-0x501F_FFFF	motherboard, 0 WS; the card may hold it off with B26/B30
32-bit card space	0x5020_0000-0x5FFF_FFFF	nEXP_ACK

Rules for nEXP_ACK:

- Assert it only while your own select is asserted.
- Once asserted, hold it until your select negates. Never pulse it.
- Read data must be valid at the connector no later than one clock after you assert it. The simplest rule is to drive the data first.
- Release it promptly: actively drive it high, or rely on the 1K pull-up (P2).
- Acknowledge within 10 us at most. The bus timeout fires after 224-240 CPU clocks (18.7 us at 12 MHz, 14 us at 16 MHz).

Drive read data (EXP_D) and nEXP_DATA_EN from 5 V CMOS outputs. The board's data transceivers U54-U57 are SN74AHC245, whose inputs need about 3.5 V for a high (TheoryOfOperations 13.2).

1.7 BUSCTRL (U58) - rev 30

Edits to plds/BUSCTRLV2/BUSCTRL2.PLD, in source order.

B1. Header and history

```
Revision 30;
//      VERSION 30 - CS1 (docs/changes.md): pin 90 nDEV_WAIT becomes nDEV_ACK; nDEV
//                   address-only; VMEM window granted here at 0 WS, with card hold
//                   bus-timeout BERR; refresh may start during non-DRAM cycles.
```

B2. Pins 90, 28 and 29, replacing Pin[90] = nDEV_WAIT;, Pin[28] = EXP_BC_CPLD1; and Pin[29] = EXP_BC_CPLD2;:

```
Pin[90] = nDEV_ACK;        // From DEVMAP pin 6. 0 = the selected device may finish
Pin[28] = nEXP_VMEM_BUSY;  // Connector B26 (was EXP_BC_CPLD1). Card holds it low w
Pin[29] = nEXP_VMEM_GNT;   // Connector B30 (was EXP_BC_CPLD2). 0 = the current CPU
```

Delete the two constant assignments EXP_BC_CPLD1 = 'b'0; and EXP_BC_CPLD2 = 'b'0;.

B3. Region decodes: replace the three `DEV8BIT_REGION_ACCESS` / `DEV16BIT_...` / `DEV32BIT_...` definitions with:

```
/* Address-only region decodes. CPU-space cycles never use 0x3-0x5 (IACK addresses
   coprocessor and BKPT addresses are 0x0002xxxx/0x0000xxxx), so FC is not needed h
DEV8_ADDR  = !CPU_A31 & !CPU_A30 &  CPU_A29 &  CPU_A28;                      // 0x
DEV16_ADDR = !CPU_A31 &  CPU_A30 & !CPU_A29 & !CPU_A28;                      // 0x
DEV32_ADDR = !CPU_A31 &  CPU_A30 & !CPU_A29 &  CPU_A28;                      // 0x
VMEM_ADDR  = DEV32_ADDR & !CPU_A27 & !CPU_A26 & !CPU_A25 & !CPU_A24
                        & !CPU_A23 & !CPU_A22 & !CPU_A21;                    // 0x

DEV8BIT_REGION_ACCESS  = DEV8_ADDR  & !nCPU_AS;
DEV16BIT_REGION_ACCESS = DEV16_ADDR & !nCPU_AS;
DEV32BIT_REGION_ACCESS = DEV32_ADDR & !nCPU_AS;
```

B4. DSACK: replace both `nCPU_DSACK0` / `nCPU_DSACK1` equations, including the `nDEV_WAIT` terms:

```
DEV8_DONE  = DEV8BIT_REGION_ACCESS  & !nDEV_ACK;
DEV16_DONE = DEV16BIT_REGION_ACCESS & !nDEV_ACK;
/* No !VMEM_ADDR term: DEVMAP never acknowledges the VMEM window, and spelling it o
   product terms per DSACK. */
DEV32_DONE = (DEV32BIT_REGION_ACCESS & !nDEV_ACK)                    // card space:
           # (DEV32BIT_REGION_ACCESS & !nEXP_VMEM_GNT);              // VMEM window

/* Terms that are not timing-critical, or never pair with DSACK1, get their own nod
   device acknowledge and the VMEM grant fit in DSACK0's own macrocell (B7). */
PINNODE = DSACK0_SLOW;
DSACK0_SLOW = DRAM_REGION_ACCESS_WAITSTATE # BOOTFLASH_REGION_ACCESS # INTERRUPT_AC

nCPU_DSACK0 = !( DSACK0_SLOW # SRAM_REGION_ACCESS # DEV8_DONE # DEV32_DONE );
nCPU_DSACK1 = !( DRAM_REGION_ACCESS_WAITSTATE # SRAM_REGION_ACCESS # BIGFLASH_REGIO
               # DEV16_DONE # DEV32_DONE );
```

Update the comment block above the DSACK equations, which still describes `nDEV_WAIT`.

B4a. VMEM grant (new, next to the DSACK equations):

```
/* VMEM window hold-off (docs/changes.md 1.4). The grant is decided on rising CPU_C
   the start of S2, then once per clock until the card's busy line is high. Once gr
   through DEV32_DONE, DSACK) stays asserted until AS negates. The card samples nEX
   next rising edge to learn whether the CPU owns the memory. Registered pin, so GN
   10 ns after the edge; DSACK follows one level later. */
nEXP_VMEM_GNT.d  = !( VMEM_ADDR & !nCPU_AS & (nEXP_VMEM_BUSY # !nEXP_VMEM_GNT) );
nEXP_VMEM_GNT.ck = CPU_CLK;
nEXP_VMEM_GNT.ap = nCPU_AS # !nSYS_RESET;     // negated between cycles and during
```

This is the same registered, preset-to-high pattern as nREFRESH_CAS.

B4b. No-card build (used for CS1, decision 9, until the VGA card exists). With no expansion card fitted, BUSCTRL can ignore B26 altogether, and then P3 is not needed. Make two changes to B4a:

```
/* No-card build: the hold-off is disabled and B26 is ignored, so no pull-up is nee
   hold-off for a VGA card, put nEXP_VMEM_BUSY back in place of 'b'1 and fit P3 at
nEXP_VMEM_GNT.d  = !( VMEM_ADDR & !nCPU_AS & ('b'1 # !nEXP_VMEM_GNT) );
```

Also add & nEXP_VMEM_BUSY to the dummy TESTP3 term, so pin 28 stays assigned as an input. The pin keeper holds it at a steady level, and since the logic never looks at it, the level does not matter. A card plugged in later cannot fight a BUSCTRL output.

Everything else is unchanged: VMEM cycles always get the grant on the S2 edge and run at 0 wait states, and nEXP_VMEM_GNT still works for the logic analyser. **Before plugging in a card that drives B26, rebuild with B4a and fit P3.**

B5. Bus timeout: add the counter, and replace nCPU_BERR = !UNUSED_REGION_ACCESS;:

```
/* Bus timeout: counts CPU_CLK/16 rising edges while AS is asserted; BERR after 15
   (224-240 CPU clocks). Catches cycles nobody acknowledges: 0x6/0x7 region, empty
   BKPT, and coprocessor cycles with no FPU fitted (the CPU then takes an F-line ex
   VMEM cycles are excluded: a card hold-off may legitimately last longer, and the
PINNODE = [BTO0..3];
[BTO0..3].ck = CPU_CLK_div16;
[BTO0..3].ar = nCPU_AS # VMEM_ADDR # !nSYS_RESET;
BTO_FULL = BTO3 & BTO2 & BTO1 & BTO0;
BTO0.d = BTO0 $ !BTO_FULL;
BTO1.d = BTO1 $ (BTO0 & !BTO_FULL);
BTO2.d = BTO2 $ (BTO1 & BTO0 & !BTO_FULL);
BTO3.d = BTO3 $ (BTO2 & BTO1 & BTO0 & !BTO_FULL);

nCPU_BERR = !( UNUSED_REGION_ACCESS # (BTO_FULL & !nCPU_AS) );
```

B6. Region selects to DEVMAP: replace nDEV8BITCS = !DEV8BIT_REGION_ACCESS; and its 16- and 32-bit companions:

```
/* Address only: DEVMAP qualifies these with its own nCPU_AS input (CS1 wire). They
   while the address changes between cycles; nothing else on the board uses them. *
nDEV8BITCS  = !DEV8_ADDR;
nDEV16BITCS = !DEV16_ADDR;
nDEV32BITCS = !DEV32_ADDR;
```

B7. Fitting DSACK into one macrocell each (done, 2026-10-18). The first rev 30 fit split both DSACKs into XXL_ helper nodes, which put the device acknowledge two levels deep. Three changes fix it:

1. Drop `!VMEM_ADDR` from `DEV32_DONE` (B4; seven product terms).
2. Drop `!FC_CPU_SPACE` from `SRAM_REGION_ACCESS` and `BIGFLASH_REGION_ACCESS`. No CPU-space address falls in 0x1xxx_xxxx or 0x08xx_xxxx. Boot flash keeps it, because coprocessor and BKPT addresses are 0x0000xxxx-0x0002xxxx.
3. The `DSACK0_SLOW` node (B4).

The node matters as much as the trims: with the trims alone, DSACK1 fit in one macrocell but DSACK0's device terms stayed in a helper. On 32-bit cycles that left about one CPLD level between the two DSACKs, against the 68030's skew limit #31A of 10 ns (20 MHz grade), 7 ns (25) or 5 ns (33). Cascade logic (`PROPERTY ATMEL {cascade_logic = on};`) was tried too: with these fixed pin positions the design no longer fits ("Placement fail").

Result: both DSACKs are single macrocells with 5 product terms each. SRAM, the device acknowledge and the VMEM grant are one level after their inputs on both. Boot flash, interrupt acknowledge and the DRAM wait-state flop go through `DSACK0_SLOW`. The design uses 79 of 128 logic cells and 34 flip-flops, and every pin stays where it was.

B8. Refresh during long cycles: replace the `REFRESH_ACTIVE.d` line, and add a sentence to the comment block above it:

```
/* CS1: a pending refresh may also start during a bus cycle that cannot be a DRAM c
   or CPU space), so long VMEM hold-offs and slow card cycles cannot starve refresh
   is still never cut into: for those the request waits for AS high, as before. */
REFRESH_ACTIVE.d = (REFRESH_REQUESTED & (nCPU_AS # !CPU_A31 # FC_CPU_SPACE)) # REFR
```

This is safe because `DRAM_REGION_ACCESS` (`!FC_CPU_SPACE & !nCPU_AS & CPU_A31 & !REFRESH_ACTIVE`) is already false during such a cycle. If the next cycle is a DRAM cycle, it waits for the refresh to finish, exactly as it does today when a refresh starts just before it. The address and FC are stable whenever AS is low, and the rising edges that sample them are half a clock away from any AS transition.

Unchanged: `nCPU_CIIN` (it now uses the new region terms, with the same meaning), the DRAM access logic, SRAM, flash, FPU select, reset, and the dividers. Apart from the B7 FC trim, SRAM and big-flash decoding is unchanged. Fitted size: 79 of 128 logic cells (rev 29 used 77). Pins 28 and 29 change use; no new pins.

1.8 DEVMAP (U59) - rev 22

Most of the logic changes, so this is a full replacement for everything after the header properties in `plds/DEVMAPV2/DEVMAP2.PLD`. The pin numbers are unchanged except pin 35. The address decodes are buried nodes (`PINNODE`) on purpose. They depend only on the address, which settles long before AS, so every select and every 0-wait-state acknowledge is a single product term on AS: one level.

```
//      VERSION 22 - CS1 (docs/changes.md): device acknowledge (nDEV_ACK) replaces
//                   nCPU_AS on pin 35 (rework wire, was TESTP0); BUSCTRL rev 30 re
//                   address-only; per-device wait-state counter; interim IDE write

/* Inputs */
Pin[52,53,54,55,56,57,58,60,61,63,64,65,67,68,69,70,71,72,75,76] = [CPU_A27..8];
Pin[100] = CPU_RW;
Pin[1]   = nDEV8BITCS;       // from BUSCTRL: address-only 0x3xxx xxxx decode (no A
Pin[2]   = nDEV16BITCS;      // 0x4xxx xxxx
Pin[5]   = nDEV32BITCS;      // 0x5xxx xxxx
Pin[35]  = nCPU_AS;          // CS1 rework wire (U70 pin 4 -> J28 pin 3)
Pin[87]  = CPU_CLK;
Pin[88]  = nSYS_RESET;
Pin[32]  = nEXP_ACK;         // connector A26 (was nEXP_DEV_WAIT): card pulls low w
Pin[33]  = nEXP_CACHE_CTRL;

/* Outputs */
Pin[6]   = nDEV_ACK;         // to BUSCTRL pin 90 (was nDEV_WAIT): 0 = the selected
Pin[29]  = nEXP_DEV8BITCS;
Pin[30]  = nEXP_DEV16BITCS;
Pin[31]  = nEXP_DEV32BITCS;
Pin[93]  = SYS_RESET_POS;
Pin[25]  = nUART_CSA;
Pin[24]  = nUART_CSB;
Pin[28]  = nPTC_CS;
Pin[50]  = nIDE1_CSA;
Pin[49]  = nIDE1_CSB;
Pin[48]  = nIDE2_CSA;
Pin[47]  = nIDE2_CSB;
Pin[46]  = nIDE_WR;
Pin[45]  = nIDE_BUFFER_EN1;
Pin[44]  = nIDE_BUFFER_EN2;
Pin[13]  = nINTCTRL_CS;
Pin[14]  = nRTC_CS;
Pin[23]  = nEXP_VMEM_CS;
Pin[22]  = nEXP_VIO_CS;
Pin[17]  = RTC_A_SEL;
Pin[16]  = RTC_D_SEL;
Pin[96]  = nSEVENSEGA_WR;
Pin[97]  = nSEVENSEGB_WR;
Pin[98]  = nSTATUSLED_WR;
Pin[7]   = EXP_DM_CPLD1;
Pin[8]   = EXP_DM_CPLD2;
Pin[21]  = EXTRA_DM_INT1;
Pin[27]  = EXTRA_DM_INT2;
Pin[83]  = EXTRA_BC_DM1;
Pin[84]  = EXTRA_BC_DM2;
Pin[36,37] = [TESTP1..2];    // J28 pins 2 and 1 (J28 pin 3 is now nCPU_AS)
Pin[92,99] = EXPANSION_EXTRA1, EXPANSION_EXTRA2;
Pin[19]  = UNUSED1;
Pin[20]  = UNUSED2;
```

```
PINNODE = [WIN0..8];
PINNODE = ONB, VMW, IDEWIN, FAST8_ADDR, FAST16_ADDR, CARD_ADDR8;
PINNODE = SLOW_ACK, nINTCTRL_CS_delay;
PINNODE = [T1..3];

FIELD field_CPU_ADDRESS = [CPU_A27..8];

/* Address-only window decodes (settle before AS). Same window numbers in the 8- an
WIN0 = field_CPU_ADDRESS:['h'00000XX];     // UART A        | IDE1 CS0
WIN1 = field_CPU_ADDRESS:['h'00001XX];     // UART B        | IDE1 CS1
WIN2 = field_CPU_ADDRESS:['h'00002XX];     // INTCTRL       | IDE2 CS0
WIN3 = field_CPU_ADDRESS:['h'00003XX];     // RTC           | IDE2 CS1
WIN4 = field_CPU_ADDRESS:['h'00004XX];     //               | PTC
WIN5 = field_CPU_ADDRESS:['h'00005XX];     // VIO (card)
WIN6 = field_CPU_ADDRESS:['h'00006XX];     // 7-segment A
WIN7 = field_CPU_ADDRESS:['h'00007XX];     // 7-segment B
WIN8 = field_CPU_ADDRESS:['h'00008XX];     // status LEDs
ONB  = field_CPU_ADDRESS:['h'0XXXXXX];     // A27-A24 all 0: on-board area of the 8
VMW  = !CPU_A27 & !CPU_A26 & !CPU_A25 & !CPU_A24 & !CPU_A23 & !CPU_A22 & !CPU_A21;
IDEWIN      = WIN0 # WIN1 # WIN2 # WIN3;
FAST8_ADDR  = ONB & !WIN3 & !WIN5;         // 8-bit on-board, 0 WS: UARTs, INTCTRL,
FAST16_ADDR = ONB & !IDEWIN;               // 16-bit on-board, 0 WS: PTC, empty win
CARD_ADDR8  = !ONB # WIN5;                 // 8-bit card space, including the VIO w

/* AS-qualified regions: one level after AS */
DEV8  = !nDEV8BITCS  & !nCPU_AS;
DEV16 = !nDEV16BITCS & !nCPU_AS;
DEV32 = !nDEV32BITCS & !nCPU_AS;

nEXP_DEV8BITCS  = !DEV8;
nEXP_DEV16BITCS = !DEV16;
nEXP_DEV32BITCS = !DEV32;

/* Device selects */
nUART_CSA       = !(DEV8 & WIN0);
nUART_CSB       = !(DEV8 & WIN1);
nINTCTRL_CS_pre = !(DEV8 & WIN2);
nRTC_CS         = !(DEV8 & WIN3);
nEXP_VIO_CS     = !(DEV8 & WIN5);
nSEVENSEGA_WR   = !(DEV8 & WIN6);
nSEVENSEGB_WR   = !(DEV8 & WIN7);
nSTATUSLED_WR   = !(DEV8 & WIN8);
nIDE1_CSA       = !(DEV16 & WIN0);
nIDE1_CSB       = !(DEV16 & WIN1);
nIDE2_CSA       = !(DEV16 & WIN2);
nIDE2_CSB       = !(DEV16 & WIN3);
nPTC_CS         = !(DEV16 & WIN4);
nEXP_VMEM_CS    = !(DEV32 & VMW);
nIDE_BUFFER_EN1 = !(DEV16 & (WIN0 # WIN1));
nIDE_BUFFER_EN2 = !(DEV16 & (WIN2 # WIN3));

/* Interrupt controller write pulse (unchanged) */
```

```
/* Interrupt-controller write pulse (unchanged) */
nINTCTRL_CS_delay.d  = 'b'1;
nINTCTRL_CS_delay.ck = !CPU_CLK;
nINTCTRL_CS_delay.ar = nINTCTRL_CS_pre # CPU_RW;
nINTCTRL_CS = nINTCTRL_CS_pre # nINTCTRL_CS_delay;

/* RTC address/data multiplexing (unchanged) */
RTC_A_SEL.d  = 'b'1;
RTC_A_SEL.ck = !CPU_CLK;
RTC_A_SEL.ar = nRTC_CS # !nSYS_RESET;
RTC_D_SEL = !RTC_A_SEL;

SYS_RESET_POS = !nSYS_RESET;

/* Timing counter, thermometer code: Tn is set by the n-th falling CPU_CLK edge aft
   and all of them clear as AS negates. T1 is set by the edge that ends S2, the CPU
   sample, so a device acknowledged by Tn gets n wait states. A decode such as Tm &
   only one input per edge, so unlike a binary count it cannot glitch. CS2 extends
[T1..3].ck = !CPU_CLK;
[T1..3].ar = nCPU_AS # !nSYS_RESET;
T1.d = 'b'1;
T2.d = T1;
T3.d = T2;

/* Acknowledge. Slow devices and cards go through the SLOW_ACK node, so the 0-wait-
   stay one level after AS. To change a device's wait states, change its Tn term. *
SLOW_ACK = (DEV8  & WIN3   & T2)                          // RTC: 2 wait states
         # (DEV16 & IDEWIN & T3)                          // IDE: 3 wait states (i
         # (((DEV8 & CARD_ADDR8) # (DEV16 & !ONB) # (DEV32 & !VMW)) & !nEXP_ACK);

nDEV_ACK = !( (DEV8 & FAST8_ADDR) # (DEV16 & FAST16_ADDR) # SLOW_ACK );

/* Interim IDE write strobe, both ports: low from the 1st to the 3rd falling edge (
   a falling-edge register, so it cannot glitch; at each edge it sees the count fro
   edge. It ends a clock before the acknowledge, while CS and the data buffers are
   CS2 replaces it. */
nIDE_WR.d  = !(DEV16 & IDEWIN & !CPU_RW & !T2);
nIDE_WR.ck = !CPU_CLK;
nIDE_WR.ap = nCPU_AS # !nSYS_RESET;

/* Unused and reserved pins (as before) */
EXP_DM_CPLD1 = 'b'0;
EXP_DM_CPLD2 = 'b'0;
EXPANSION_EXTRA2 = nSYS_RESET & CPU_RW & nEXP_ACK & nEXP_CACHE_CTRL;
EXPANSION_EXTRA1 = EXTRA_DM_INT1 & EXTRA_DM_INT2 & EXTRA_BC_DM1 & EXTRA_BC_DM2 & EX
UNUSED1 = UNUSED2;

/* Test points on J28 */
TESTP1 = nDEV_ACK;     // J28 pin 2
TESTP2 = T1;           // J28 pin 1
```

Removed compared with rev 20: `IDE_WR_SIG_A`, `IDE_WR_SIG_RESET`, `IDE_WR_SIG`, `nIDE_WAITSTATE_DELAY`, `..._RESETw1`, `..._RESETw2`, `nDEV_WAIT`, and `TESTP0`. After compiling, check that `nDEV_ACK` and the device selects use only the `WINn/ONB/VMW/FASTx` nodes plus single product terms on `nCPU_AS`.

Interim IDE timing this gives, before CS2:

	12 MHz	**16 MHz**
Cycle length	6 clocks, 500 ns	6 clocks, 375 ns
/IOW low (counts 1-2)	167 ns	125 ns
/IOR (`nCPU_RD`, S3 to the end of the cycle)	about 330 ns	about 250 ns

That is already about ATA PIO mode 0-1 at 12 MHz and mode 1-2 at 16 MHz, with CS and data held after /IOW.

1.9 INTCTRL and PTC

No change in CS1.

1.10 Software notes (BIOS and OS)

- Device addresses and register layouts do not change.
- Accesses that used to hang now take a **bus error** after the timeout (14-20 us): 0x6000_0000-0x7FFF_FFFF, `BKPT`, and FPU instructions with no FPU fitted. The last one becomes an F-line exception, so a software FPU emulator can be installed.
- Accesses to **empty card space** (for example 0x3100_0000, the VIO window, or 0x5020_0000 with no card) now bus-error after the timeout instead of returning floating data. Probing code needs a bus-error handler in place.
- Empty **on-board** windows (for example 0x3000_0400) still acknowledge at once, as today.
- The RTC now runs with 2 wait states and IDE with 3. They are slower per access, but within spec.
- VMEM accesses (0x5000_0000-0x501F_FFFF) can stall while a VGA card is busy, but they never bus-error. Without a card they run at 0 wait states and read floating data.

1.11 Programming, wiring and test

1. Build BUSCTRL rev 30 and DEVMAP rev 22. Keep `REV29_BUSCTRL2-CPURD.jed` and `REV21-DEVMAP2-2WS.jed` for rollback.
2. Program **both** CPLDs in the same session; each needs the other's change. The board will not run between this step and step 4.
3. Power off and fit the P1 wire (U70 pin 4 to J28 pin 3). P3 is not needed: CS1 uses the no-card build (B4b, decision 9).

4. Power on and boot. Checks, with the logic analyser on U70 (nCPU_AS pin 4, nCPU_DSACK0/1 pins 8/9, CPU_CLK pin 3), J28 (pin 2 nDEV_ACK, pin 1 T1) and U69 (pin 3 nEXP_VMEM_BUSY, pin 2 nEXP_VMEM_GNT):

Test	Expected
UART / console	AS low about 2 clocks (0 WS); 3 at 16 MHz is acceptable
RTC time read and NVRAM write/read	2 wait states, data correct
IDE IDENTIFY on IDE1 and on IDE2	3 wait states, nIDE_WR low for 2 clocks inside the cycle (U4 pin 17 = D2)
Read of 0x3100_0000	bus error after about 15-20 us
Read of 0x6000_0000	bus error (it hangs today)
Read of 0x5000_0000	0 wait states, floating data; nEXP_VMEM_GNT low from S2 to the end of each access
Hold-off (B4a build only): run a loop reading 0x5000_0000, then link U69 pin 3 to pin 20 (GND) for a few seconds	the CPU stops inside the read with no bus error, and carries on when the link is removed
Refresh during a stall (B4a build only): fill DRAM with a pattern, hold U69 pin 3 low for 10 s during a VMEM read, release, check the pattern	pattern intact (with the old refresh condition the DRAM would lose it)
No-card build (B4b): link U69 pin 3 to GND while reading 0x5000_0000	no effect: reads still run at 0 wait states
Normal OS boot and run	no change

1.12 Rollback

Power off and **remove the P1 wire first**: the old DEVMAP drives pin 35. Then program the two saved JEDs. To keep the wire fitted while rolling back, rebuild the old DEVMAP with pin 35 declared as an unused input. The P3 pull-up can stay.

1.13 Next PCB items from CS1

All next-PCB items from every change set are consolidated, with pin assignments, in NextPCB.md.

- Route nCPU_AS to a DEVMAP dedicated input (pin 89 or 90; both are unconnected today) and give TESTP0 back its header pin.
- Name connector A26 nEXP_ACK with a 1K pull-up. Name B26 nEXP_VMEM_BUSY with a 4.7K pull-up, and B30 nEXP_VMEM_GNT.

- Optionally route A31-A28 to DEVMAP, so it can decode regions without BUSCTRL's help.
- Change the schematic value of the DSACK pull-ups R51/R52 from 10K to 1K, to match the boards.

1.14 Decisions

#	Question	Status
1	VMEM fast-window size	**Decided 2026-10-18:** 2 MB, 0x5000_0000-0x501F_FFFF
2	RTC wait states	**Decided 2026-10-18:** 2. DS becomes about 3 clocks long, closer to the DS12887 data sheet.
3	Empty on-board windows	**Decided for now (2026-10-18):** acknowledge at once, as today. May be revisited (bus error instead).
4	Hold-off for the dual-use video memory	**Decided 2026-10-18:** busy/grant pair on B26/B30, decided on clock edges, part of CS1 (1.4)
5	Longest hold-off a card may use	Open. Proposed: at most one scan line (about 32 us) at a time. Optional: a much longer VMEM watchdog in BUSCTRL (about 4 more macrocells), so a stuck card bus-errors instead of hanging the machine.
6	Bus-timeout length	Open. 224-240 clocks proposed.
7	P2 (R10 to 1K)	Open: now, or with the first card.
8	AS to DEVMAP: rework wire P1, or relay AS through BUSCTRL on the spare `EXTRA_BC_DM2`	**Decided 2026-10-18: the wire (P1).** A relay passes through one BUSCTRL level, exactly like today's AS-qualified region selects. That gives no gain: the fast on-board devices would still take 1 wait state at 16 MHz.
9	Build BUSCTRL rev 30 without the VMEM hold-off (B4b) until the VGA card exists, deferring P3	**Decided 2026-10-18: use B4b.** P3 is fitted, and B4a enabled, when the VGA card is ready.

CS2 - IDE

CS1 already gives IDE real wait states, covers both ports and puts /IOW inside the cycle, with interim timing (1.8). CS2 finishes the job:

- ATA-compliant strobe and cycle timing on both ports (PIO mode 0), plus a faster alias for the 16-bit Data register (PIO mode 2);
- an /IOR window shaped by DEVMAP, and /CS held after the strobes;
- IDE interrupts;
- the two pull-downs each port is missing;
- a reset pulse long enough for the drives.

It builds on CS1: the AS wire, nDEV_ACK, and DEVMAP's timing counter.

2.1 Every IDE problem, and where it is fixed

#	Problem	Effect	Fixed by
1	BUSCTRL acknowledges the 16-bit region before DEVMAP's wait request can arrive (TheoryOfOperations 12.5)	every IDE cycle runs with 0 wait states	CS1 (device acknowledge)
2	/IOW rises after /CS0 and the data buffers have turned off	writes lost, or written to the wrong register	CS1 interim; final timing in D3
3	IDE2 never gets /IOW or wait states	IDE2 writes do nothing	CS1 (all four selects); D3
4	The wait request leaks into the next cycle	inconsistent timing on back-to-back accesses	CS1 (old flops removed)
5	/IOR is about one clock long, starts too soon after /CS (ATA t1) and ends together with /CS and the address (t9)	slow drives return stale data; a Data-register read may be lost or counted twice	D3 (/IOR window, /CS hold), B9
6	Strobe and cycle times are far below PIO mode 0, which register access needs (290 ns strobes, 600 ns cycles)	marginal register access, worst on older drives	2.2 base timing
7	Strobes decoded from a binary counter can glitch; CS1's first draft had one on /IOW	a glitch on /IOW is an extra write	CS1, already corrected: thermometer counter and registered /IOW
8	INTRQ is read as active low, its mask bits are forced off, and the line floats whenever the drive lets go of it	no IDE interrupts	I1, I2, P5
9	DD7 has no pull-down	with no drive fitted, Status can read as busy forever	P6
10	The software reset gives the drives about one clock of /RESET; ATA needs 25 us	drives may not reset on a soft reboot	B10
11	CSEL (pin 28) is not grounded; IORDY (27) and DASP (39) are not connected	cable select cannot work; no flow control; no activity LED	notes in 2.3; next PCB (2.10)
12	/IOR, /IOW and /CS drive the cable with no series termination	ringing can double-clock a strobe	check with a scope (2.8); next PCB

#	Problem	Effect	Fixed by
13	The IDE data transceivers are SN74AHC245 (per the netlist), with CMOS input thresholds; ATA drives only promise 2.4 V for a high	read data from drives with TTL-level or 3.3 V outputs has little or no noise margin	P7 (proposed)

2.2 Timing

Two speeds. In PIO mode 0 the 8-bit registers need 290 ns strobes and a 600 ns cycle, and the strobe limit stays at 290 ns even in modes 1 and 2. The 16-bit Data register, which carries all the sector data, needs only 100 ns strobes and a 240 ns cycle in mode 2. DEVMAP cannot see A7-A5, so it cannot tell the registers apart. Instead the IDE registers appear twice, and address bit A11 picks the timing:

Window	IDE1	IDE2	Timing	Use
Base	0x4000_0000-0x4000_01FF	0x4000_0200-0x4000_03FF	PIO mode 0, 10-clock cycle	everything; the only window for the 8-bit registers
Fast	0x4000_0800-0x4000_09FF	0x4000_0A00-0x4000_0BFF	PIO mode 2, 5-clock cycle	the Data register (0x4000_0800, 0x4000_0A00) on drives that report mode 2 or better

The register layout inside each window is unchanged (TheoryOfOperations 12.2). The fast window used to be empty on-board space.

Counts. DEVMAP's timing counter numbers the falling CPU_CLK edges after AS asserts; edge 1 ends S2. The strobes assert and deassert on those edges. An acknowledge at Tn ends the cycle with n wait states, and the CPU latches read data on edge n + 2.

	/IOW low	/IOR low	Acknowledge	Cycle
Base	edges 2 to 7 (5 clocks)	edges 2 to 9 (7 clocks)	T7	10 clocks
Fast	edges 1 to 3 (2 clocks)	edges 1 to 4 (3 clocks)	T2	5 clocks

The rules behind the numbers:

- each strobe starts on edge 1 or 2, late enough after /CS for t1;
- /IOW ends at least a clock before the end of the cycle, so data and /CS stay valid after it (t4, t9);
- /IOR ends exactly on the data-latch edge;
- the whole cycle meets t0.

Worst case at the drive, in ns. CPLD paths use the -10 figures. Delays through U39/U42 cancel where both signals pass through them.

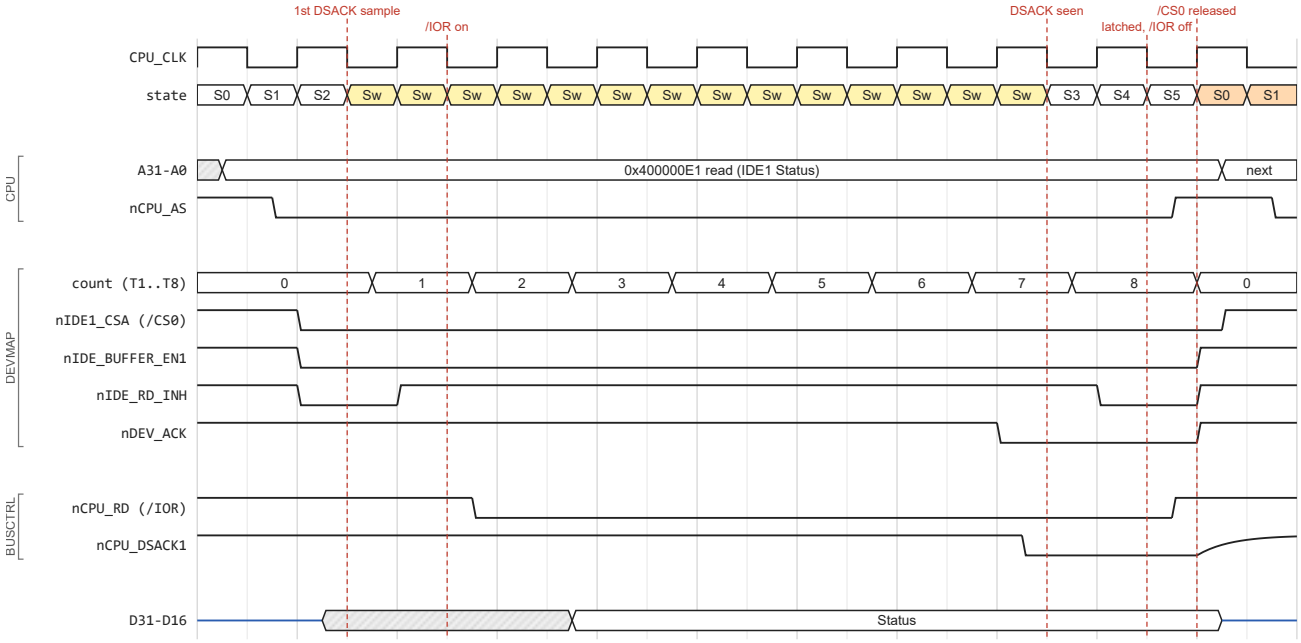
Parameter	Base, 12 MHz	Base, 16 MHz	Mode 0 limit	Fast, 12 MHz	Fast, 16 MHz	Mode 2 limit
t0 cycle	833	625	600	417	312	240
t1 /CS and address to strobe	138	96	70	54	33	30
t2 /IOW width	417	312	290 (8-bit)	167	125	100 (16-bit)
t2 /IOR width	583	437	290 (8-bit)	250	187	100 (16-bit)
t4 write data hold after /IOW	157	116	30	74	53	15
t9 /CS hold after /IOW	155	113	20	71	50	10
t9 /CS hold after /IOR	38	27	20	38	27	10
t9 address hold after /IOR	33	22	20	33	22	10
Read access time left for the drive	556	410	240 (t2 - t5)	223	160	80

The counts are set for 16 MHz, so at 12 MHz everything is simply slower. Above 16 MHz the base counts must grow; t0 is the first limit to fail.

How a read ends.

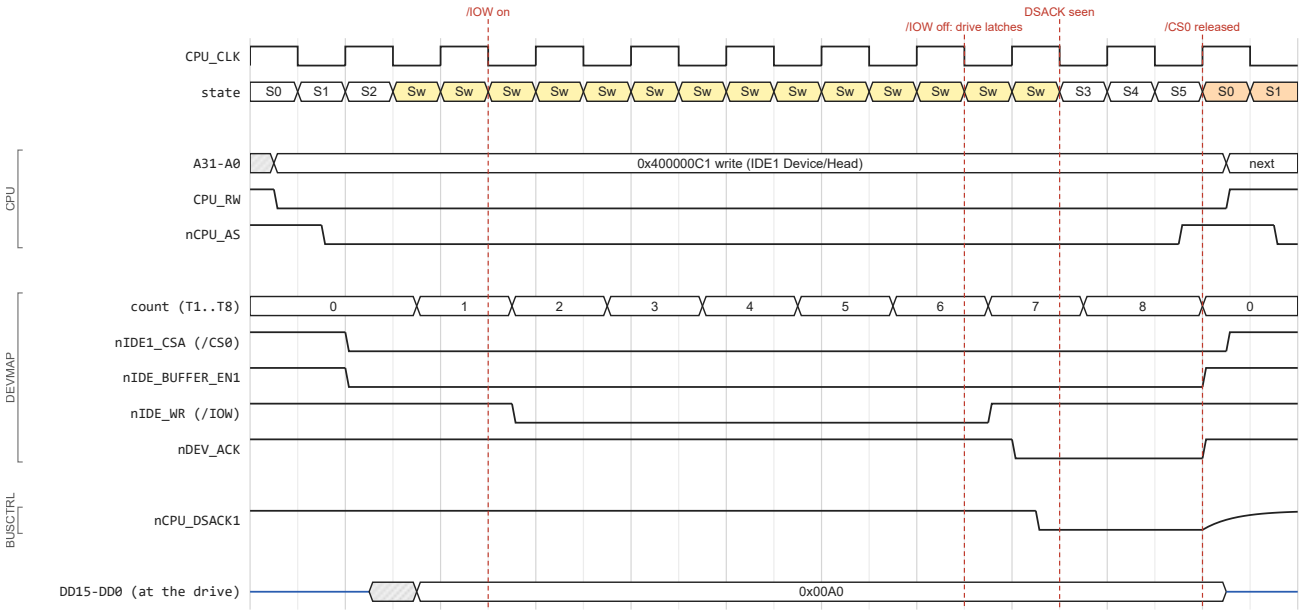
- /IOR rises on the edge where the CPU latches the data, so the drive still drives the bus through the latch.
- /CS stays asserted until the next rising edge (the `IDEH` flops in D3): at least 27 ns more at 16 MHz.
- The address is the tighter limit. The CPU changes it as the next cycle starts, half a clock after the latch: 22 ns at 16 MHz, against 20 ns for mode 0. Today /IOR, /CS and the address all change together.
- The cost is on the CPU side. After /IOR rises, a drive only guarantees its data for 5 ns (t6). That covers the 68030's input latch in practice, but not the letter of spec #29 if DS negates late.

PROPOSED (changes.md CS2): IDE register read, base window (PIO mode 0), 10-clock cycle

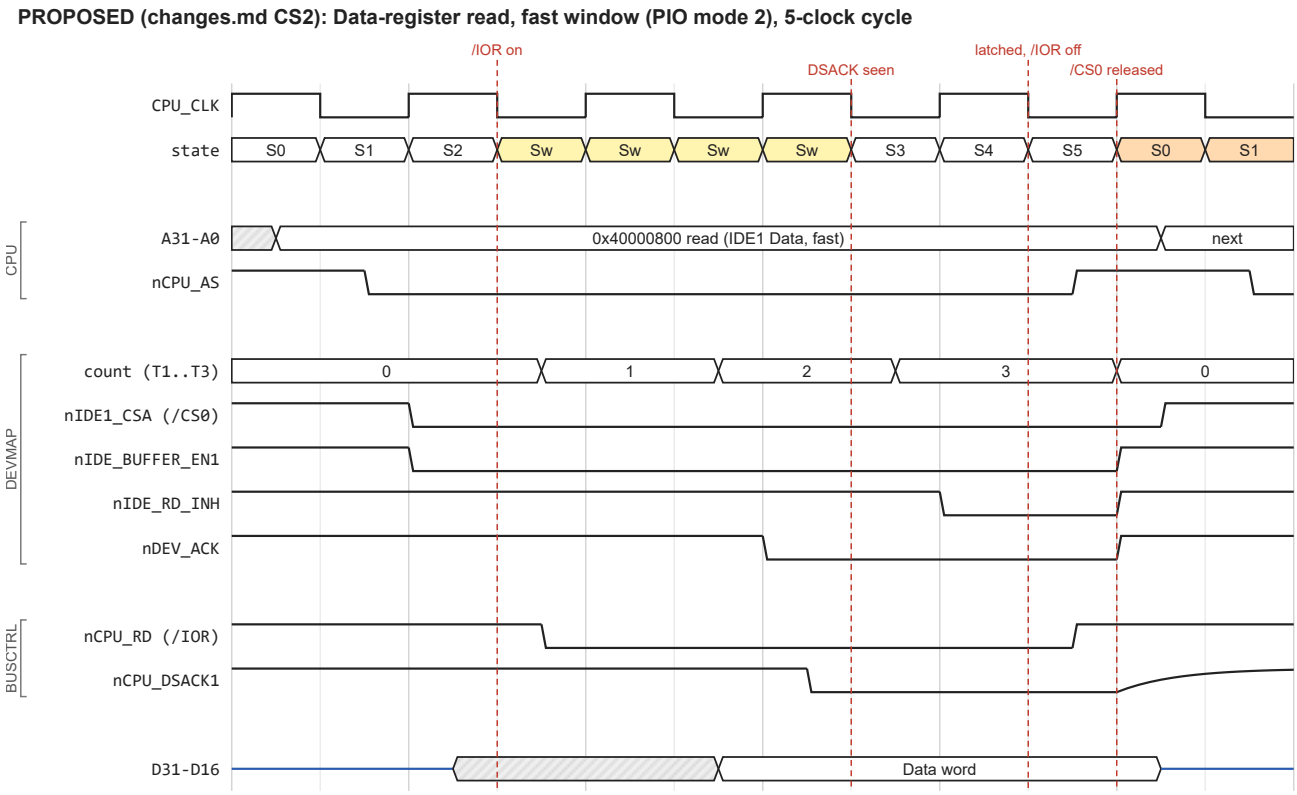


Proposed. Drawn at 16 MHz (1 slot = 15.6 ns). DEVMAP's thermometer counter steps on each falling edge after AS. nIDE_RD_INH is low for the first clock and again from T8, and BUSCTRL's nCPU_RD flop samples it on falling edges, so /IOR is low from edge 2 to edge 9: 437 ns at 16 MHz. DEVMAP acknowledges at T7, so the CPU sees DSACK1 on edge 8 and latches the data on edge 9, the same edge that ends /IOR. The IDEH flop keeps /CS0 asserted until the next rising edge, after /IOR has risen, while the data transceivers turn off with AS. A fast-window read (0x4000_0800) uses the same logic with T3 and T2 for a 5-clock cycle.

PROPOSED (changes.md CS2): IDE register write, base window (PIO mode 0), 10-clock cycle



Proposed. Drawn at 16 MHz (1 slot = 15.6 ns). nIDE_WR is a falling-edge register fed by the counter, so it cannot glitch: low from edge 2 to edge 7, 312 ns. The drive latches the data on the rising edge of /IOW, two clocks before AS negates, with /CS0 and the data still valid (t9 113 ns and t4 116 ns at 16 MHz). nIDE_RD_INH stays high in write cycles, so nCPU_RD never asserts. IDE2 uses the same strobe; only the selects differ. A fast-window write has /IOW from edge 1 to edge 3 and a 5-clock cycle.



Proposed. Drawn at 16 MHz (1 slot = 15.6 ns). A11 = 1 selects the fast timing: /IOR (nCPU_RD) is low from edge 1 to edge 4 (187 ns), DEVMAP acknowledges at T2, and the cycle takes 5 clocks (312 ns). That meets PIO mode 2 for 16-bit transfers at 12 and 16 MHz. Use the fast window for the Data register only: the 8-bit registers need the base window's 290 ns strobes. A fast write has /IOW from edge 1 to edge 3.

2.3 Physical changes

#	Change	When	Notes
P5	**10K pull-down on INTRQ**: J4 pin 31 to J4 pin 30 (GND), and J6 pin 31 to J6 pin 30	before enabling IDE interrupts (I1, I2)	A drive tri-states INTRQ while nIEN is set or the other drive is selected. Without the pull-down the line floats, and INTCTRL's pin keeper holds whatever level it last had.
P6	**10K pull-down on DD7**: J4 pin 3 to J4 pin 2 (GND), and J6 pin 3 to J6 pin 2	with CS2	The usual host-side pull-down. With no drive fitted, Status then reads with BSY clear, so probe code cannot hang on an empty port.
P7	*Proposed:* **replace U37, U38, U40, U41 (SN74AHC245) with SN74AHCT245**	with CS2, or as soon as a drive misreads	Same SOIC-20 package and pinout. AHCT has TTL input thresholds (a high from 2.0 V), which matches what ATA drives promise (2.4 V). The AHC parts want about 3.5 V, which drives with 3.3 V or TTL outputs may not reach. Check which parts are fitted first; if they are already AHCT, P7 is done.

P5 and P6 are single resistors between neighbouring pins on the back of the connectors. All the pins in this table were checked against the netlist (2026-10-18): neither INTRQ nor DD7 has a resistor today, and pins 2 and 30 are GND.

No wire is needed for /IOR. DEVMAP sends the /IOR window to BUSCTRL over the spare net `EXTRA_BC_DM1` (DEVMAP pin 83 to BUSCTRL pin 81), which already exists and is unused. It also appears on logic header U5 (D15, pin 4). The alternative is a rework: lift pin 8 of U39 and U42 (their /IOR inputs) off `nCPU_RD`, and drive them from a DEVMAP pin. That gives the same timing with one CPLD fewer in the path, so it is listed for the next PCB instead (2.10).

Notes, no change needed now:

- Jumper the drives as master and slave. CSEL (pin 28) is not grounded, so a drive set to cable select becomes the slave (device 1).
- IORDY (pin 27) is not connected. The base timing does not need it, and neither does mode 2 for the Data register.
- Power on pin 20, for DOMs and CF adapters, comes from jumper J5 (IDE1) or J7 (IDE2).

2.4 DEVMAP (U59) - rev 23

D1. The old IDE code (rev 20) that goes away. It is shown here without its comments. CS1's rev 22 already removed it; it is repeated here so the old and new versions can be compared.

```
/* 16 bit devices */
nIDE1_CSA = nDEV16BITCS # !(field_CPU_ADDRESS:['h'00000XX]);
nIDE1_CSB = nDEV16BITCS # !(field_CPU_ADDRESS:['h'00001XX]);
nIDE2_CSA = nDEV16BITCS # !(field_CPU_ADDRESS:['h'00002XX]);
nIDE2_CSB = nDEV16BITCS # !(field_CPU_ADDRESS:['h'00003XX]);

nIDE_BUFFER_EN1 = (nIDE1_CSA & nIDE1_CSB);
nIDE_BUFFER_EN2 = (nIDE2_CSA & nIDE2_CSB);

nIDE_WR = CPU_RW # !IDE_WR_SIG;

IDE_WR_SIG_A.d = 'b'1;
IDE_WR_SIG_A.ck = !CPU_CLK;
IDE_WR_SIG_A.ar = (nIDE1_CSA & nIDE1_CSB) # !nSYS_RESET;

IDE_WR_SIG_RESET.d = IDE_WR_SIG;
IDE_WR_SIG_RESET.ck = CPU_CLK;
IDE_WR_SIG_RESET.ar = (!nDEV_WAIT) # !nSYS_RESET;

IDE_WR_SIG.d = 'b'1;
IDE_WR_SIG.ck = CPU_CLK;
IDE_WR_SIG.ar = (!IDE_WR_SIG_A) # IDE_WR_SIG_RESET # !nSYS_RESET;

nIDE_WAITSTATE_DELAY.d = 'b'1;
nIDE_WAITSTATE_DELAY.ck = !(nIDE1_CSA & nIDE1_CSB);
nIDE_WAITSTATE_DELAY.ar = nIDE_WAITSTATE_DELAY_RESETw2 # !nSYS_RESET;

nIDE_WAITSTATE_DELAY_RESETw1.d = 'b'1;
nIDE_WAITSTATE_DELAY_RESETw1.ck = !CPU_CLK;
nIDE_WAITSTATE_DELAY_RESETw1.ar = !nIDE_WAITSTATE_DELAY # !nSYS_RESET;

nIDE_WAITSTATE_DELAY_RESETw2.d =  nIDE_WAITSTATE_DELAY_RESETw1;
nIDE_WAITSTATE_DELAY_RESETw2.ck = !CPU_CLK;
nIDE_WAITSTATE_DELAY_RESETw2.ar = !nIDE_WAITSTATE_DELAY # !nSYS_RESET;

nDEV_WAIT = (!nIDE_WAITSTATE_DELAY) & nEXP_DEV_WAIT;

TESTP0 = nDEV_WAIT;
TESTP1 = nIDE_WAITSTATE_DELAY;
TESTP2 = nIDE_WAITSTATE_DELAY_RESETw2;
```

D2. What replaces each part

Rev 20	What was wrong	Rev 23
nIDEx_CSx from BUSCTRL's AS-qualified region select	/CS two CPLDs after AS, and it ends with AS	IDE1A..IDE2B from DEV16 (direct AS), held by the IDEH0..3 flops until the rising edge after AS negates

Rev 20	What was wrong	Rev 23
nIDE_BUFFER_ENx from the CS outputs	would inherit the /CS hold	built from the AS-qualified selects only, so the transceivers let go of the CPU bus as AS negates
IDE_WR_SIG_A, IDE_WR_SIG, IDE_WR_SIG_RESET	/IOW ends after /CS and the data; IDE1 only	nIDE_WR: one registered output from the timing counter, both ports
nIDE_WAITSTATE_DELAY, ..._RESETw1/w2, nDEV_WAIT	the wait request arrives too late and leaks into the next cycle; IDE1 only	counter T1..T8 and the IDE terms in SLOW_ACK, through CS1's nDEV_ACK
none: /IOR is BUSCTRL's nCPU_RD	/IOR about one clock; poor t1 and t9	nIDE_RD_INH on pin 83, which BUSCTRL folds into nCPU_RD (B9)
TESTP0..2		TESTP1 = nDEV_ACK, TESTP2 = T1 (J28 pin 3 is nCPU_AS since CS1)

D3. Rev 23 listing. This is a complete replacement for everything after the header properties, like CS1's rev 22. Compared with rev 22, the changes are:

- the version note and pin 83;
- the IDEW and IDEH nodes (IDEWIN is gone);
- the counter, extended to T8;
- the IDE selects and /IOW;
- the new nIDE_RD_INH;
- the IDE terms in SLOW_ACK, plus FAST16_ADDR and EXPANSION_EXTRA1.

```
//      VERSION 23 - CS2 (docs/changes.md): IDE timing. Counter T1..T8; IDE selects
//                   (A11) and held to the rising edge after AS; registered /IOW; /
//                   BUSCTRL on pin 83 (nIDE_RD_INH).

/* Inputs */
Pin[52,53,54,55,56,57,58,60,61,63,64,65,67,68,69,70,71,72,75,76] = [CPU_A27..8];
Pin[100] = CPU_RW;
Pin[1]   = nDEV8BITCS;       // from BUSCTRL: address-only 0x3xxx xxxx decode (no A
Pin[2]   = nDEV16BITCS;      // 0x4xxx xxxx
Pin[5]   = nDEV32BITCS;      // 0x5xxx xxxx
Pin[35]  = nCPU_AS;          // CS1 rework wire (U70 pin 4 -> J28 pin 3)
Pin[87]  = CPU_CLK;
Pin[88]  = nSYS_RESET;
Pin[32]  = nEXP_ACK;         // connector A26: card pulls low when its cycle may en
Pin[33]  = nEXP_CACHE_CTRL;

/* Outputs */
Pin[6]   = nDEV_ACK;         // to BUSCTRL pin 90: 0 = the selected device may fini
Pin[83]  = nIDE_RD_INH;      // to BUSCTRL pin 81 (net EXTRA_BC_DM1): 0 = keep /IOR
Pin[29]  = nEXP_DEV8BITCS;
Pin[30]  = nEXP_DEV16BITCS;
Pin[31]  = nEXP_DEV32BITCS;
Pin[93]  = SYS_RESET_POS;
Pin[25]  = nUART_CSA;
Pin[24]  = nUART_CSB;
Pin[28]  = nPTC_CS;
Pin[50]  = nIDE1_CSA;
Pin[49]  = nIDE1_CSB;
Pin[48]  = nIDE2_CSA;
Pin[47]  = nIDE2_CSB;
Pin[46]  = nIDE_WR;
Pin[45]  = nIDE_BUFFER_EN1;
Pin[44]  = nIDE_BUFFER_EN2;
Pin[13]  = nINTCTRL_CS;
Pin[14]  = nRTC_CS;
Pin[23]  = nEXP_VMEM_CS;
Pin[22]  = nEXP_VIO_CS;
Pin[17]  = RTC_A_SEL;
Pin[16]  = RTC_D_SEL;
Pin[96]  = nSEVENSEGA_WR;
Pin[97]  = nSEVENSEGB_WR;
Pin[98]  = nSTATUSLED_WR;
Pin[7]   = EXP_DM_CPLD1;
Pin[8]   = EXP_DM_CPLD2;
Pin[21]  = EXTRA_DM_INT1;
Pin[27]  = EXTRA_DM_INT2;
Pin[84]  = EXTRA_BC_DM2;
Pin[36,37] = [TESTP1..2];    // J28 pins 2 and 1 (J28 pin 3 is nCPU_AS)
Pin[92,99] = EXPANSION_EXTRA1, EXPANSION_EXTRA2;
Pin[19]  = UNUSED1;
Pin[20]  = UNUSED2;
```

```
PINNODE = [WIN0..8];
PINNODE = ONB, VMW, IDEW, FAST8_ADDR, FAST16_ADDR, CARD_ADDR8;
PINNODE = SLOW_ACK, nINTCTRL_CS_delay;
PINNODE = [T1..8];
PINNODE = [IDEH0..3];

FIELD field_CPU_ADDRESS = [CPU_A27..8];

/* Address-only window decodes (settle before AS). WIN4 is the PTC in the 16-bit re
   others are the 8-bit devices. */
WIN0 = field_CPU_ADDRESS:['h'00000XX];     // UART A
WIN1 = field_CPU_ADDRESS:['h'00001XX];     // UART B
WIN2 = field_CPU_ADDRESS:['h'00002XX];     // INTCTRL
WIN3 = field_CPU_ADDRESS:['h'00003XX];     // RTC
WIN4 = field_CPU_ADDRESS:['h'00004XX];     // PTC
WIN5 = field_CPU_ADDRESS:['h'00005XX];     // VIO (card)
WIN6 = field_CPU_ADDRESS:['h'00006XX];     // 7-segment A
WIN7 = field_CPU_ADDRESS:['h'00007XX];     // 7-segment B
WIN8 = field_CPU_ADDRESS:['h'00008XX];     // status LEDs
ONB  = field_CPU_ADDRESS:['h'0XXXXXX];     // A27-A24 all 0: on-board area of the 8
VMW  = !CPU_A27 & !CPU_A26 & !CPU_A25 & !CPU_A24 & !CPU_A23 & !CPU_A22 & !CPU_A21;

/* IDE: 0x4000_0000-0x4000_03FF (base timing) and the same registers at 0x4000_0800
   (A11 = 1: fast timing, for the Data register). A9 picks the port and A8 picks CS
IDEW = !CPU_A27 & !CPU_A26 & !CPU_A25 & !CPU_A24 & !CPU_A23 & !CPU_A22 & !CPU_A21 &
     & !CPU_A19 & !CPU_A18 & !CPU_A17 & !CPU_A16 & !CPU_A15 & !CPU_A14 & !CPU_A13 &
     & !CPU_A10;
IDE_FAST = CPU_A11;

FAST8_ADDR  = ONB & !WIN3 & !WIN5;         // 8-bit on-board, 0 WS: UARTs, INTCTRL,
FAST16_ADDR = ONB & !IDEW;                 // 16-bit on-board, 0 WS: PTC, empty win
CARD_ADDR8  = !ONB # WIN5;                 // 8-bit card space, including the VIO w

/* AS-qualified regions: one level after AS */
DEV8  = !nDEV8BITCS  & !nCPU_AS;
DEV16 = !nDEV16BITCS & !nCPU_AS;
DEV32 = !nDEV32BITCS & !nCPU_AS;

nEXP_DEV8BITCS  = !DEV8;
nEXP_DEV16BITCS = !DEV16;
nEXP_DEV32BITCS = !DEV32;

/* Device selects other than IDE (as rev 22) */
nUART_CSA       = !(DEV8 & WIN0);
nUART_CSB       = !(DEV8 & WIN1);
nINTCTRL_CS_pre = !(DEV8 & WIN2);
nRTC_CS         = !(DEV8 & WIN3);
nEXP_VIO_CS     = !(DEV8 & WIN5);
nSEVENSEGA_WR   = !(DEV8 & WIN6);
nSEVENSEGB_WR   = !(DEV8 & WIN7);
nSTATUSLED_WR   = !(DEV8 & WIN8);
nPTC_CS         = !(DEV16 & WIN4);
```

```
nRTC_CS         = !(DEV16 & WIN4);
nEXP_VMEM_CS    = !(DEV32 & VMW);

/* Interrupt-controller write pulse (unchanged) */
nINTCTRL_CS_delay.d  = 'b'1;
nINTCTRL_CS_delay.ck = !CPU_CLK;
nINTCTRL_CS_delay.ar = nINTCTRL_CS_pre # CPU_RW;
nINTCTRL_CS = nINTCTRL_CS_pre # nINTCTRL_CS_delay;

/* RTC address/data multiplexing (unchanged) */
RTC_A_SEL.d  = 'b'1;
RTC_A_SEL.ck = !CPU_CLK;
RTC_A_SEL.ar = nRTC_CS # !nSYS_RESET;
RTC_D_SEL = !RTC_A_SEL;

SYS_RESET_POS = !nSYS_RESET;

/* Timing counter, thermometer code: Tn is set by the n-th falling CPU_CLK edge aft
   and all of them clear as AS negates. A device acknowledged by Tn gets n wait sta
   the form Tm & !Tn change one input per edge, so they cannot glitch. */
[T1..8].ck = !CPU_CLK;
[T1..8].ar = nCPU_AS # !nSYS_RESET;
T1.d = 'b'1;
T2.d = T1;
T3.d = T2;
T4.d = T3;
T5.d = T4;
T6.d = T5;
T7.d = T6;
T8.d = T7;

/* IDE selects. They assert one level after AS. Each IDEH flop holds its select unt
   rising edge after AS negates, so /CS stays valid after /IOR rises (ATA t9). The
   the AS-qualified selects only, so they let go of the CPU bus as AS negates. */
IDE_CYC = DEV16 & IDEW;
IDE1A = IDE_CYC & !CPU_A9 & !CPU_A8;       // IDE1 /CS0
IDE1B = IDE_CYC & !CPU_A9 &  CPU_A8;       // IDE1 /CS1
IDE2A = IDE_CYC &  CPU_A9 & !CPU_A8;       // IDE2 /CS0
IDE2B = IDE_CYC &  CPU_A9 &  CPU_A8;       // IDE2 /CS1

[IDEH0..3].ck = CPU_CLK;
[IDEH0..3].ar = !nSYS_RESET;
IDEH0.d = IDE1A;
IDEH1.d = IDE1B;
IDEH2.d = IDE2A;
IDEH3.d = IDE2B;

nIDE1_CSA = !(IDE1A # IDEH0);
nIDE1_CSB = !(IDE1B # IDEH1);
nIDE2_CSA = !(IDE2A # IDEH2);
nIDE2_CSB = !(IDE2B # IDEH3);
```

```
nIDE_BUFFER_EN1 = !(IDE1A # IDE1B);
nIDE_BUFFER_EN2 = !(IDE2A # IDE2B);

/* /IOW, both ports. A falling-edge register, so it cannot glitch; at each edge it
   from before the edge. Base: low from edge 2 to edge 7. Fast: from edge 1 to edge
nIDE_WR.d  = !( IDE_CYC & !CPU_RW & ( (!IDE_FAST & T1 & !T6) # (IDE_FAST & !T2) ) )
nIDE_WR.ck = !CPU_CLK;
nIDE_WR.ap = nCPU_AS # !nSYS_RESET;

/* /IOR window. /IOR is BUSCTRL's nCPU_RD, a falling-edge register that can only as
   where this signal is high, and that sees the count from the edge before. Base: /
   edge 2 to edge 9. Fast: from edge 1 to edge 4. Both end on the edge where the CP
   data. High in every other cycle, so nCPU_RD behaves as before for all other devi
nIDE_RD_INH = !( IDE_CYC & CPU_RW & ( (!IDE_FAST & (!T1 # T8)) # (IDE_FAST & T3) )

/* Acknowledge. Slow devices and cards go through the SLOW_ACK node, so the 0-wait-
   stay one level after AS. IDE timing lives in three places that must change toget
   T terms here, in nIDE_WR and in nIDE_RD_INH (docs/changes.md 2.2). */
SLOW_ACK = (DEV8  & WIN3 & T2)                                      // RTC: 2 wait
         # (IDE_CYC & !IDE_FAST & T7)                               // IDE base: 7
         # (IDE_CYC &  IDE_FAST & T2)                               // IDE fast: 2
         # (((DEV8 & CARD_ADDR8) # (DEV16 & !ONB) # (DEV32 & !VMW)) & !nEXP_ACK);

nDEV_ACK = !( (DEV8 & FAST8_ADDR) # (DEV16 & FAST16_ADDR) # SLOW_ACK );

/* Unused and reserved pins */
EXP_DM_CPLD1 = 'b'0;
EXP_DM_CPLD2 = 'b'0;
EXPANSION_EXTRA2 = nSYS_RESET & CPU_RW & nEXP_ACK & nEXP_CACHE_CTRL;
EXPANSION_EXTRA1 = EXTRA_DM_INT1 & EXTRA_DM_INT2 & EXTRA_BC_DM2 & EXP_DM_CPLD1 & EX
UNUSED1 = UNUSED2;

/* Test points on J28 */
TESTP1 = nDEV_ACK;     // J28 pin 2
TESTP2 = T1;           // J28 pin 1
```

After compiling, check in the fit report that nIDE_WR is a registered output, and that nDEV_ACK's 0-wait-state terms still depend only on the address nodes plus nCPU_AS. Resource estimate: about 60 of 128 macrocells.

2.5 BUSCTRL (U58) - rev 31

Edits on top of rev 30 (1.7).

B9. /IOR window from DEVMAP. Old (rev 29, unchanged in rev 30):

```
Pin[81] = EXTRA_BC_DM1;

nCPU_RD.d = !CPU_RW;
nCPU_RD.ck = !CPU_CLK;
nCPU_RD.ap = nCPU_AS # !nSYS_RESET;

TESTP3 = CPU_A19 & CPU_A18 & CPU_A17 & EXTRA_BC_DM1 & EXTRA_BC_DM2 & EXTRA_BC_INT1
```

New:

```
Pin[81] = nIDE_RD_INH;     // from DEVMAP pin 83 (net EXTRA_BC_DM1): 0 = keep nCPU_

/* nCPU_RD is the read strobe: /OE for SRAM and flash, /IOR for the UARTs and both
   For IDE reads DEVMAP shapes it through nIDE_RD_INH (docs/changes.md 2.2). For ev
   nIDE_RD_INH is high and nCPU_RD is as in rev 29: low from S3 to the end of the c
nCPU_RD.d  = !(CPU_RW & nIDE_RD_INH);
nCPU_RD.ck = !CPU_CLK;
nCPU_RD.ap = nCPU_AS # !nSYS_RESET;

TESTP3 = CPU_A19 & CPU_A18 & CPU_A17 & nIDE_RD_INH & EXTRA_BC_DM2 & EXTRA_BC_INT1 &
```

`nIDE_RD_INH` has to be stable 7 ns before each falling edge. DEVMAP changes it at most about 18 ns after a falling edge (counter plus one level). At the start of a cycle it follows AS, so it can come as late as 34 ns after the edge that begins S1. Either way the margin is at least 20 ns at 16 MHz.

B10. Reset long enough for the drives. Old:

```
nSYS_RESET.d = 'b'1;
nSYS_RESET.ar = !nSYS_RESET_IN # RESET_CONTROL_IN;
nSYS_RESET.ck = !CPU_CLK;
```

New:

```
/* Reset stretch (CS2): after a reset request ends, hold nSYS_RESET low for 31 more
   CPU_CLK/16 (480-497 CPU clocks: 30-31 us at 16 MHz, 40-41 us at 12 MHz). The sof
   (INTCTRL 0x30000202 bit 1) used to last about one clock, against the 25 us ATA d
   /RESET. The power-on reset gets the same extra time. */
PINNODE = [RST0..4];
[RST0..4].ck = CPU_CLK_div16;
[RST0..4].ar = !nSYS_RESET_IN # RESET_CONTROL_IN;
RST_DONE = RST4 & RST3 & RST2 & RST1 & RST0;
RST0.d = RST0 $ !RST_DONE;
RST1.d = RST1 $ (RST0 & !RST_DONE);
RST2.d = RST2 $ (RST1 & RST0 & !RST_DONE);
RST3.d = RST3 $ (RST2 & RST1 & RST0 & !RST_DONE);
RST4.d = RST4 $ (RST3 & RST2 & RST1 & RST0 & !RST_DONE);

nSYS_RESET.d  = RST_DONE;
nSYS_RESET.ar = !nSYS_RESET_IN # RESET_CONTROL_IN;
nSYS_RESET.ck = !CPU_CLK;
```

The release still happens on a falling CPU_CLK edge, as before. The software-reset request clears itself as soon as nSYS_RESET falls (INTCTRL resets its own register), and that is what starts the count. B10 is independent of the rest of CS2 and also fixes TheoryOfOperations finding #7.

B11. Header: `Revision 31;` and

```
//      VERSION 31 - CS2 (docs/changes.md): nCPU_RD gated by DEVMAP's nIDE_RD_INH f
//                   reset stretched to about 490 clocks.
```

Resources: 5 more macrocells (about 87 of 128). Pin 81 changes from a dummy input to a used input; no new pins.

2.6 INTCTRL (U60) - rev 17

I1. Let software enable the IDE interrupts. Old:

```
[INT_MSK11].d = 'b'0; //CPU_D3.io;  // IDE1
[INT_MSK10].d = 'b'0; //CPU_D2.io;  // IDE2
```

New:

```
[INT_MSK11].d = CPU_D3.io;  // IDE1
[INT_MSK10].d = CPU_D2.io;  // IDE2
```

I2. INTRQ is active high. Old:

```
  INT5B_REQUEST = !vINT5B & INT_MSK11;        // INT5B comes from the IDE #1 interfac
  INT5C_REQUEST = !vINT5C & INT_MSK10;        // INT5C comes from the IDE #2 interfac
```

New:

```
  INT5B_REQUEST = vINT5B & INT_MSK11;         // IDE1 INTRQ (J4 pin 31): active high,
  INT5C_REQUEST = vINT5C & INT_MSK10;         // IDE2 INTRQ (J6 pin 31): active high,
```

I3. Header: `Revision 17;` and `// VERSION 17 - CS2 (docs/changes.md): IDE interrupts: INTRQ active high, mask bits writable.`

The masks reset to 0, so nothing changes until software sets them. Fit P5 before setting them.

2.7 Software notes

- Register addresses do not change. The fast Data-register addresses are 0x4000_0800 (IDE1) and 0x4000_0A00 (IDE2). A long-word move still works: the CPU splits it into two 16-bit cycles, and both hit the Data register.
- Use the fast window only for the Data register, and only on drives whose IDENTIFY data shows PIO mode 2 or better: word 51 bits 15-8 at least 2, or word 53 bit 1 set and word 64 non-zero. Keep every 8-bit register on the base window.
- Interrupts (after P5 and INTCTRL rev 17):
    - set bit 3 (IDE1) or bit 2 (IDE2) of the mask register at 0x3000_0200;
    - clear nIEN (Device Control bit 1) on the drive;
    - in the handler, read Status (not Alternate Status), which clears INTRQ.

  The vectors are 0xD2 (IDE1) and 0xD4 (IDE2), level 5.
- Reset: the software reset now holds /RESET for 30-41 us. After any /RESET, wait 2 ms, then poll BSY; a drive may take up to 31 s to spin up. To reset just the drives, use SRST (Device Control bit 2) instead of a system reset.
- Byte order is unchanged (TheoryOfOperations 12.3): swap the bytes in each word for PC-compatible sector data.
- IORDY: the board ignores IORDY (CS3 can add it for the base window). After IDENTIFY, check word 49:
    - If bit 10 is set ("IORDY may be disabled"), send SET FEATURES with Features = 03h and Sector Count = 01h ("PIO default mode, IORDY disabled"), so the drive never relies on it.
    - If bit 11 is set but bit 10 is clear, the drive may stretch cycles with IORDY. Use the base window for its Data register too, and consider CS3 if it misbehaves.
- With P6, an empty port reads BSY clear, so probing cannot hang on it. Check for a drive by writing and reading back the Sector Count register.

2.8 Programming, wiring and test

1. CS1 must be in place and verified.
2. Build DEVMAP rev 23 and BUSCTRL rev 31, and program both in the same session: `nIDE_RD_INH` runs between them. INTCTRL rev 17 can come later. Keep the rev 22 and rev 30 JEDs for rollback.
3. Fit P6 now. Fit P5 before enabling IDE interrupts.
4. Checks, with the logic analyser on:
    - U70: `nCPU_AS` pin 4, `CPU_RW` pin 5, `nCPU_DSACK1` pin 9, `nSYS_RESET` pin 13, `CPU_CLK` pin 3;
    - U4: `nIDE1_CSA` pin 15, `nIDE1_BUFFER_EN` pin 16, `nIDE_WR` pin 17;
    - U5: `nIDE_RD_INH` pin 4;
    - J4 itself: /IOR pin 25, /IOW pin 23, /CS0 pin 37.

Test	Expected
Write 0xA0 to Device/Head (0x4000_00C1)	AS low for 9 clocks (a 10-clock cycle); /IOW low from the 2nd to the 7th falling edge; /CS0 rises after AS
Read Status (0x4000_00E1)	/IOR low from the 2nd to the 9th falling edge, and rising before /CS0; `nIDE_RD_INH` low in the first clock and in the last
IDENTIFY DEVICE on IDE1 and IDE2, through the base window	the same data as on a PC, allowing for byte order
Read IDENTIFY again through the fast window (0x4000_0800, 0x4000_0A00)	5-clock cycles; identical data
Write sectors, read them back and compare, both ports, many sectors	no mismatches
Scope at the drive end of the cable: /IOR and /IOW	one clean edge each way, with no ringing back across 0.8-2.0 V. If they ring, use a shorter cable and plan series resistors (2.10).
Software reset (write 0x02 to 0x3000_0202)	`nSYS_RESET` low for about 490 clocks; drives ready again afterwards
After P5 and INTCTRL rev 17: READ SECTOR with nIEN = 0 and the mask bit set	one level-5 interrupt per sector, vector 0xD2 (IDE1) or 0xD4 (IDE2)

2.9 Rollback

Program the CS1 JEDs (DEVMAP rev 22, BUSCTRL rev 30) and INTCTRL rev 16. P5 and P6 can stay: with rev 16 the IDE interrupt masks are forced off, and the DD7 pull-down is harmless.

2.10 Next PCB items from CS2

Consolidated, with pin assignments, in NextPCB.md.

- IORDY (pin 27) to a DEVMAP input, with a 1K pull-up, so slow devices can stretch cycles.
- IDE /IOR from its own DEVMAP pin, instead of `nCPU_RD` (frees `EXTRA_BC_DM1`).
- 10K pull-downs on INTRQ (pin 31) and DD7 (pin 3), on both ports.
- Series resistors (22-33 ohm) at the drivers for /IOR, /IOW, /CS0, /CS1 and A2-A0, and ideally the data lines.
- CSEL (pin 28) to GND through a jumper, for cable select.
- DASP (pin 39) to an activity LED, with a pull-up.
- SN74AHCT245 for the IDE data transceivers (P7) and for the expansion data transceivers U54-U57.

2.11 Decisions

#	Question	Status
1	Base timing: PIO mode 0 for every access	**Agreed 2026-10-18**
2	Fast alias for the Data register at base + 0x800 (PIO mode 2)	**Agreed 2026-10-18**
3	/IOR through BUSCTRL and `EXTRA_BC_DM1` (no rework), rather than lifting U39/U42 pin 8	**Agreed 2026-10-18**
4	Stretch every reset to about 490 clocks (B10; affects the whole board)	**Agreed 2026-10-18**
5	Fit P5 and P6 now	**Agreed 2026-10-18**
6	P7: SN74AHCT245 in the IDE data path	Proposed (added after the netlist check)

CS3 - IDE IORDY (optional)

Do this only if some cards still fail after CS2, **and** the check in 3.4 shows them pulling IORDY low. CS3 builds on CS2 (DEVMAP rev 23).

3.1 Why it is optional

A drive that needs more time pulls IORDY (pin 27) low within 35 ns of a strobe starting (ATA tA), and may hold it low for up to 1.25 us (tB). The host must keep the strobe asserted until IORDY is released. The host has to honour IORDY in PIO modes 3 and 4. In modes 0-2 it is optional, because a drive that meets those modes needs no flow control.

CS2 runs every register access at mode 0 (/IOW 312 ns, /IOR 437 ns at 16 MHz) and the Data-register alias at mode 2, so a compliant drive never needs IORDY. Today's one-clock strobes are another matter, which is why IORDY may be part of the current failures. The remaining risk after CS2 is a drive that insists on

IORDY in modes 0-2 and holds it longer than the base strobes. The software step in 2.7 (SET FEATURES "IORDY disabled") covers every drive that allows it.

The fast window cannot use IORDY. Its acknowledge comes at the second falling edge, before IORDY could be seen, and DSACK can never be withdrawn. CS3 therefore acts on the base window only, which is fine: mode 2 does not require IORDY.

3.2 Physical change

#	Change	When	Notes
P8	**IORDY to DEVMAP**: wire J4 pin 27 to J6 pin 27, and from there to J28 pin 1 (DEVMAP pin 37, today TESTP2). Add a **1K pull-up** from J4 pin 27 to J5 pin 1 (VCC, next to J4).	with CS3. **Only connect J28 pin 1 after DEVMAP rev 24 is programmed**: rev 23 drives pin 37 as an output.	1K is the ATA host value. Only the selected drive ever drives IORDY, so both ports can share the line and the pull-up. All the pins were checked in the netlist: pin 27 is unconnected on both headers, and J5 pin 1 is VCC.

3.3 DEVMAP (U59) - rev 24

Edits to rev 23 (2.4).

D4. Pin 37 becomes the IORDY input. Replace `Pin[36,37] = [TESTP1..2];` with:

```
Pin[36]  = TESTP1;           // J28 pin 2
Pin[37]  = IDE_IORDY;        // J28 pin 1 (was TESTP2): IORDY from J4/J6 pin 27, 1K
```

Delete `TESTP2 = T1;`.

D5. Sample IORDY, and hold the counter while it is low. Add the node and replace the counter block:

```
PINNODE = IORDY_S;

/* IORDY is asynchronous, so it is sampled on each falling edge. Preset on reset so
IORDY_S.d  = IDE_IORDY;
IORDY_S.ck = !CPU_CLK;
IORDY_S.ap = !nSYS_RESET;

/* IORDY hold, base window only: once a strobe has started (T2) and the drive holds
   counter stops where it is. /IOW, the /IOR window (nIDE_RD_INH) and the acknowled
   the counter, so they stretch together and BUSCTRL needs no change. T1 and T2 are
   the RTC (T2) and every other device are unaffected. */
IDE_HOLD = IDE_CYC & !IDE_FAST & T2 & !IORDY_S;

[T1..8].ck = !CPU_CLK;
[T1..8].ar = nCPU_AS # !nSYS_RESET;
T1.d = 'b'1;
T2.d = T1;
T3.d = T3 # (T2 & !IDE_HOLD);
T4.d = T4 # (T3 & !IDE_HOLD);
T5.d = T5 # (T4 & !IDE_HOLD);
T6.d = T6 # (T5 & !IDE_HOLD);
T7.d = T7 # (T6 & !IDE_HOLD);
T8.d = T8 # (T7 & !IDE_HOLD);
```

D6. Header: `// VERSION 24 - CS3 (docs/changes.md): IDE IORDY on pin 37 holds the base-window timing counter.`

Timing, base window at 16 MHz:

- /IOR or /IOW reaches the drive about 16 ns after edge 2.
- A drive that wants more time pulls IORDY low within 35 ns (tA), so DEVMAP sees it about 52 ns after edge 2. That meets the 7 ns setup before edge 3 with a few ns to spare. If it is missed, edge 4 catches it.
- The counter therefore stops at T3 or T4, well before /IOW ends (T6), the acknowledge (T7) and the end of /IOR (T8).
- When IORDY rises, the counter resumes on the next-but-one edge, so the strobe still runs 3 to 5 more clocks. ATA only requires the data to be valid when IORDY rises.
- A drive that never releases IORDY is caught by CS1's bus timeout: the access ends in a bus error after about 15 us.

Resources: one more flop, and an extra product term on T3..T8.

3.4 Check before building it

Find out first whether any of your cards actually use IORDY:

1. Fit only the 1K pull-up from P8 (J4 pin 27 to J5 pin 1), with nothing connected to J28.

2. Clip the logic analyser or a scope onto J4 pin 27.
3. Run the CS2 tests in 2.8 with each card.

If IORDY never goes low, CS3 is not needed. The pull-up can stay; it is part of P8 anyway.

3.5 Programming, test and rollback

1. Program DEVMAP rev 24 with J28 pin 1 still unconnected, then power off and fit the P8 wire.
2. Repeat the CS2 tests (2.8) on the cards that failed. Then test the hold with **no drive on J4**:
    - Run a loop reading 0x4000_00E1, and link J4 pin 27 to pin 26 (GND).
    - Each read should now stall until the bus timeout ends it with a bus error, after about 15 us.
    - With the link removed, reads complete normally again.

    The dangling pull-up only draws 5 mA through the link.
3. Rollback: remove the wire from J28 pin 1 first, then program rev 23. The pull-up can stay.

3.6 Decisions

#	Question	Status
1	Build CS3 at all	Open: decide after CS2 testing and the 3.4 check
2	IORDY on J28 pin 1 (losing the `T1` test point) for this board	Proposed

Change log

Date	Change
2026-10-18	File created. CS0 recorded (rework already on the boards). CS1 proposed: device acknowledge, AS wire to DEVMAP, VMEM fast window, bus timeout.
2026-10-18	CS1 decisions recorded: 2 MB VMEM window, RTC 2 wait states, empty on-board windows unchanged. Added the VMEM hold-off (`nEXP_VMEM_BUSY` on B26, `nEXP_VMEM_GNT` on B30, BUSCTRL B4a), excluded VMEM from the bus timeout, and let refresh start during non-DRAM cycles (B8). P3 is now required for CS1. P4 dropped: the DSACK pull-ups are already 1K.
2026-10-18	Hold-off decided. CS1 DEVMAP: binary wait-state counter replaced by a thermometer counter, and the interim /IOW made a register (the decoded version could glitch). CS2 written: PIO mode 0 base timing plus a mode 2 Data-register alias, /IOR window through BUSCTRL, /CS hold, IDE interrupts, INTRQ and DD7 pull-downs (P5, P6), reset stretch.

Date	Change
2026-10-18	CS2 agreed as proposed: all five decisions in 2.11.
2026-10-18	Checked against the netlist (`schematic/RoscoeMainboardV2Final.net`): every CPLD pin assignment and every rework point in CS1 and CS2 matches. Proposed P7 (SN74AHCT245 in the IDE data path), after the netlist showed SN74AHC245 there. Added the 5 V CMOS rule for card read data.
2026-10-18	CS1 decision 8: keep the P1 wire rather than relaying AS through BUSCTRL.
2026-10-18	CS0 corrected from the underside photos: H12 and F13 go to one spare pull-up (R44/R45 via J23) and RTC /IRQ (U33 pin 19) to the other; /CBACK (J1) is not wired. Added 0.3, a reset-to-UART wire whose purpose is still to be confirmed.
2026-10-18	CS0 0.3 identified: a filter capacitor from the reset-button net to VCC (J301 pin 2), added because the button was noisy.
2026-10-18	CS3 written (optional): IORDY to DEVMAP pin 37 via J28 pin 1 plus a 1K pull-up (P8), and a counter hold in DEVMAP rev 24. Added the SET FEATURES "IORDY disabled" step to the CS2 software notes. Next-PCB routing consolidated in NextPCB.md.
2026-10-18	CS1: added the no-card BUSCTRL build (B4b), which ignores B26 so P3 can wait for the VGA card (decision 9).
2026-10-18	Decision 9 decided: CS1 uses the no-card build (B4b); P3 is deferred to the VGA card.
2026-10-18	CS1 implemented in the sources: `BUSCTRL2.PLD` rev 30 (B1-B8 with B4b: `VMEM_READY = 'b'1`) and `DEVMAP2.PLD` rev 22 (the 1.8 listing). Awaiting WinCUPL compile and fit review.
2026-10-18	CS1 fit review. DEVMAP rev 22: 36/128 cells, pins kept, fast acknowledges one level after AS. BUSCTRL rev 30: DSACK trimmed (B7: no `!VMEM_ADDR` in `DEV32_DONE`, no FC on SRAM/big flash, `DSACK0_SLOW` node), so both DSACKs fit in one macrocell; 79/128 cells, pins kept. Cascade logic was tried and does not fit with the fixed pinout.
2026-10-18	WinCUPL would not open `DEVMAP2.PLD` rev 22 ("doesn't exist. New File assumed"). Its editor's symbol parser crashes on a line holding a `field` reference, then `;`, then =, which the ONB line had in its comment. Reworded that comment in the source and in the 1.8 and D3 listings. Added `plds/check-wincupl-open.ps1`, which runs the same parser over .PLD files. DEVMAP rev 22 now compiles in WinCUPL with the same fit (36/128).
2026-10-18	BUSCTRL rev 30 compiled in WinCUPL: 79/128 cells, 34 flip-flops, 223 product terms, no cascades. Both DSACKs are single macrocells with `nDEV_ACK` and `nEXP_VMEM_GNT` one level deep. Every declared pin is at its declared position in both CPLDs. DEVMAP pins 21, 27, 83 and 84 are unused, as in rev 20. Ready for 1.11 step 2.